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Probability

JEE Main Mathematics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer) 183 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

A candidate has to go to the examination centre to appear in an examination. The candidate uses only one means of transportation for the entire distance out of bus, scooter and car. The probabilities of the candidate going by bus, scooter and car, respectively, are $\frac{2}{5}$, $\frac{1}{5}$ and $\frac{2}{5}$. The probabilities that the candidate reaches late at the examination centre are $\frac{1}{5}$, $\frac{1}{3}$ and $\frac{1}{4}$ if the candidate uses bus, scooter and car, respectively. Given that the candidate reached late at the examination centre, the probability that the candidate travelled by bus is :

(A) $\frac{11}{37}$ (B) $\frac{12}{37}$ (C) $\frac{13}{37}$ (D) $\frac{14}{37}$

Open solution

Q2. JEE Main 2026 (Online) 6th April Evening Shift

A bag contains 6 blue and 6 green balls. Pairs of balls are drawn without replacement until the bag is empty. The probability that each drawn pair consists of one blue and one green ball is :

(A) $\frac{63}{925}$ (B) $\frac{17}{231}$ (C) $\frac{16}{231}$ (D) $\frac{64}{925}$

Open solution

Q3. JEE Main 2026 (Online) 6th April Morning Shift

A bag contains $(\mathrm{N}+1)$ coins -N fair coins, and one coin with 'Head' on both sides. A coin is selected at random and tossed. If the probability of getting 'Head' is $\frac{9}{16}$, then N is equal to:

(A) 5 (B) 7 (C) 8 (D) 9

Open solution

Q4. JEE Main 2026 (Online) 5th April Evening Shift

The probabilities that players A and B of a team are selected for the captaincy for a tournament are 0.6 and 0.4 , respectively. If A is selected the captain, the probability that the team wins the tournament is 0.8 and if B is selected the captain, the probability that the team wins the tournament is 0.7 . Then the probability, that the team wins the tournament, is :

(A) 0.74 (B) 0.76 (C) 0.72 (D) 0.78

Open solution

Q5. JEE Main 2026 (Online) 5th April Morning Shift

A letter is known to have arrived by post either from KANPUR or from ANANTPUR. On the envelope just two consecutive letters AN are visible. The probability, that the letter came from ANANTPUR, is:

(A) $\frac{7}{10}$ (B) $\frac{10}{17}$ (C) $\frac{12}{19}$ (D) $\frac{7}{19}$

Open solution

Q6. JEE Main 2026 (Online) 2nd April Evening Shift

A man throws a fair coin repeatedly. He gets 10 points for each head he throws and 5 points for each tail he throws. If the probability that he gets exactly 30 points is $\frac{m}{n}$, gcd $(m, n) = 1$, then m + n is equal to :

(A) 53 (B) 55 (C) 107 (D) 105

Open solution

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Q7. JEE Main 2026 (Online) 28th January Evening Shift

The probability distribution of a random variable X is given below :

X	4k	$\frac{30}{7}k$	$\frac{32}{7}k$	$\frac{34}{7}k$	$\frac{36}{7}k$	$\frac{38}{7}k$	$\frac{40}{7}k$	6k
P(X)	$\frac{2}{15}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{5}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{5}$	$\frac{1}{15}$

If E(X) = $\frac{263}{15}$, then P(X < 20) is equal to :

(A) $\frac{3}{5}$ (B) $\frac{14}{15}$ (C) $\frac{8}{15}$ (D) $\frac{11}{15}$

Open solution

Q8. JEE Main 2026 (Online) 28th January Morning Shift

A bag contains 10 balls out of which k are red and $(10-k)$ are black, where $0 \leq k \leq 10$. If three balls are drawn at random without replacement and all of them are found to be black, then the probability that the bag contains 1 red and 9 black balls is:

(A) $\frac{7}{110}$ (B) $\frac{7}{11}$ (C) $\frac{7}{55}$ (D) $\frac{14}{55}$

Open solution

Q9. JEE Main 2026 (Online) 24th January Morning Shift

From a lot containing 10 defective and 90 non-defective bulbs, 8 bulbs are selected one by one with replacement. Then the probability of getting at least 7 defective bulbs is

(A) $\frac{73}{10^8}$ (B) $\frac{67}{10^8}$ (C) $\frac{7}{10^7}$ (D) $\frac{81}{10^8}$

Open solution

Q10. JEE Main 2026 (Online) 23rd January Evening Shift

Bag A contains 9 white and 8 black balls, while bag B contains 6 white and 4 black balls. One ball is randomly picked up from the bag B and mixed up with the balls in the bag A . Then a ball is randomly drawn from the bag A . If the probability, that the ball drawn is white, is $\frac{\mathrm{p}}{\mathrm{q}}, \gcd(\mathrm{p}, \mathrm{q}) = 1$, then $\mathrm{p} + \mathrm{q}$ is equal to

(A) 24 (B) 22 (C) 23 (D) 21

Open solution

Q11. JEE Main 2026 (Online) 22nd January Morning Shift

Two distinct numbers a and b are selected at random from $1, 2, 3, \ldots, 50$. The probability, that their product ab is divisible by 3 , is

(A) $\frac{272}{1225}$ (B) $\frac{561}{1225}$ (C) $\frac{664}{1225}$ (D) $\frac{8}{25}$

Open solution

Q12. JEE Main 2026 (Online) 22nd January Morning Shift

If a random variable x has the probability distribution

x	0	1	2	3	4	5	6	7
$\mathrm{P}(x)$	0	$2\mathrm{k}$	k	$3\mathrm{k}$	$2\mathrm{k}^2$	$2\mathrm{k}$	$\mathrm{k}^2+\mathrm{k}$	$7\mathrm{k}^2$

then $\mathrm{P}(3 < x \leq 6)$ is equal to

(A) 0.34 (B) 0.64 (C) 0.22 (D) 0.33

Open solution

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Q13. JEE Main 2026 (Online) 21st January Evening Shift

A random variable X takes values 0, 1, 2, 3 with probabilities $\frac{2a+1}{30}$, $\frac{8a-1}{30}$, $\frac{4a+1}{30}$, b respectively, where $a, b \in \mathbb{R}$.

Let μ and σ respectively be the mean and standard deviation of X such that $\sigma^2 + \mu^2 = 2$.

Then $\frac{a}{b}$ is equal to:

(A) 12 (B) 60 (C) 30 (D) 3

Open solution

Q14. JEE Main 2026 (Online) 21st January Morning Shift

Let the mean and variance of 7 observations $2, 4, 10, x, 12, 14, y, x > y$, be 8 and 16 respectively. Two numbers are chosen from $\{1, 2, 3, x-4, y, 5\}$ one after another without replacement, then the probability, that the smaller number among the two chosen numbers is less than 4 , is :

(A) $\frac{4}{5}$ (B) $\frac{3}{5}$ (C) $\frac{2}{5}$ (D) $\frac{1}{3}$

Open solution

Q15. JEE Main 2025 (Online) 8th April Evening Shift

If *A* and *B* are two events such that $P(A) = 0.7$, $P(B) = 0.4$ and $P(A \cap \overline{B}) = 0.5$, where $\overline{B}$ denotes the complement of *B*, then $P\left(B \mid (A \cup \overline{B})\right)$ is equal to

(A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{1}{6}$

Open solution

Q16. JEE Main 2025 (Online) 7th April Evening Shift

A bag contains 19 unbiased coins and one coin with head on both sides. One coin drawn at random is tossed and head turns up. If the probability that the drawn coin was unbiased, is $\frac{m}{n}$, $\gcd(m, n) = 1$, then $n^2 - m^2$ is equal to :

(A) 64 (B) 80 (C) 60 (D) 72

Open solution

Q17. JEE Main 2025 (Online) 7th April Evening Shift

Let a random variable X take values 0, 1, 2, 3 with P(X=0)=P(X=1)=p, P(X=2)=P(X=3) and $E(X^2)=2E(X)$. Then the value of 8p−1 is :

(A) 2 (B) 0 (C) 3 (D) 1

Open solution

Q18. JEE Main 2025 (Online) 4th April Morning Shift

The probability, of forming a 12 persons committee from 4 engineers, 2 doctors and 10 professors containing at least 3 engineers and at least 1 doctor, is

(A) $\frac{129}{182}$ (B) $\frac{17}{26}$ (C) $\frac{19}{26}$ (D) $\frac{103}{182}$

Open solution

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Q19. JEE Main 2025 (Online) 4th April Morning Shift

A box contains 10 pens of which 3 are defective. A sample of 2 pens is drawn at random and let X denote the number of defective pens. Then the variance of X is

(A) $\frac{11}{15}$ (B) $\frac{2}{15}$ (C) $\frac{3}{5}$ (D) $\frac{28}{75}$

Open solution

Q20. JEE Main 2025 (Online) 3rd April Evening Shift

If the probability that the random variable X takes the value x is given by $P(X=x)=k(x+1)3^{-x}, x=0,1,2,3\ldots$, where k is a constant, then $P(X \geq 3)$ is equal to

(A) $\frac{1}{9}$ (B) $\frac{8}{27}$ (C) $\frac{7}{27}$ (D) $\frac{4}{9}$

Open solution

Q21. JEE Main 2025 (Online) 2nd April Evening Shift

Given three identical bags each containing 10 balls, whose colours are as follows :

	Red	Blue	Green
Bag I	3	2	5
Bag II	4	3	3
Bag III	5	1	4

A person chooses a bag at random and takes out a ball. If the ball is Red, the probability that it is from bag I is p and if the ball is Green, the probability that it is from bag III is q, then the value of $\left(\frac{1}{p}+\frac{1}{q}\right)$ is:

(A) 6 (B) 9 (C) 7 (D) 8

Open solution

Q22. JEE Main 2025 (Online) 29th January Evening Shift

Bag 1 contains 4 white balls and 5 black balls, and Bag 2 contains *n* white balls and 3 black balls. One ball is drawn randomly from Bag 1 and transferred to Bag 2. A ball is then drawn randomly from Bag 2. If the probability, that the ball drawn is white, is $\frac{29}{45}$, then *n* is equal to:

(A) 5 (B) 6 (C) 4 (D) 3

Open solution

Q23. JEE Main 2025 (Online) 28th January Evening Shift

Bag B_1 contains 6 white and 4 blue balls, Bag B_2 contains 4 white and 6 blue balls, and Bag B_3 contains 5 white and 5 blue balls. One of the bags is selected at random and a ball is drawn from it. If the ball is white, then the probability that the ball is drawn from Bag B_2 is:

(A) $\frac{2}{5}$ (B) $\frac{4}{15}$ (C) $\frac{1}{3}$ (D) $\frac{2}{3}$

Open solution

Q24. JEE Main 2025 (Online) 28th January Evening Shift

Let S be the set of all the words that can be formed by arranging all the letters of the word GARDEN. From the set S, one word is selected at random. The probability that the selected word will NOT have vowels in alphabetical order is:

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{2}{3}$

Open solution

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Q25. JEE Main 2025 (Online) 28th January Morning Shift

Two number k_1 and k_2 are randomly chosen from the set of natural numbers. Then, the probability that the value of $i^{k_1}+i^{k_2}, (i=\sqrt{-1})$ is non-zero, equals

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{2}{3}$

Open solution

Q26. JEE Main 2025 (Online) 28th January Morning Shift

Three defective oranges are accidentally mixed with seven good ones and on looking at them, it is not possible to differentiate between them. Two oranges are drawn at random from the lot. If x denotes the number of defective oranges, then the variance of x is

(A) $26/75$ (B) $14/25$ (C) $18/25$ (D) $28/75$

Open solution

Q27. JEE Main 2025 (Online) 24th January Evening Shift

Let $A=[a_{ij}]$ be a square matrix of order 2 with entries either 0 or 1 . Let E be the event that A is an invertible matrix. Then the probability $P(E)$ is :

(A) $\frac{3}{8}$ (B) $\frac{1}{8}$ (C) $\frac{3}{16}$ (D) $\frac{5}{8}$

Open solution

Q28. JEE Main 2025 (Online) 24th January Morning Shift

A and B alternately throw a pair of dice. A wins if he throws a sum of 5 before B throws a sum of 8 , and B wins if he throws a sum of 8 before A throws a sum of 5 . The probability, that A wins if A makes the first throw, is

(A) $\frac{8}{19}$ (B) $\frac{9}{19}$ (C) $\frac{8}{17}$ (D) $\frac{9}{17}$

Open solution

Q29. JEE Main 2025 (Online) 23rd January Evening Shift

A board has 16 squares as shown in the figure :



Out of these 16 squares, two squares are chosen at random. The probability that they have no side in common is :

(A) $\frac{3}{5}$ (B) $\frac{4}{5}$ (C) $\frac{23}{30}$ (D) $\frac{7}{10}$

Open solution

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Q30. JEE Main 2025 (Online) 23rd January Morning Shift

One die has two faces marked 1 , two faces marked 2 , one face marked 3 and one face marked 4 . Another die has one face marked 1 , two faces marked 2 , two faces marked 3 and one face marked 4. The probability of getting the sum of numbers to be 4 or 5 , when both the dice are thrown together, is

(A) $\frac{2}{3}$ (B) $\frac{3}{5}$ (C) $\frac{4}{9}$ (D) $\frac{1}{2}$

Open solution

Q31. JEE Main 2025 (Online) 22nd January Evening Shift

If A and B are two events such that $P(A \cap B) = 0.1$, and $P(A \mid B)$ and $P(B \mid A)$ are the roots of the equation $12x^2 - 7x + 1 = 0$, then the value of $\frac{P(\bar{A} \cup \bar{B})}{P(\bar{A} \cap \bar{B})}$ is :

(A) $\frac{4}{3}$ (B) $\frac{7}{4}$ (C) $\frac{9}{4}$ (D) $\frac{5}{3}$

Open solution

Q32. JEE Main 2025 (Online) 22nd January Morning Shift

A coin is tossed three times. Let X denote the number of times a tail follows a head. If μ and σ^2 denote the mean and variance of X, then the value of $64\left(\mu + \sigma^2\right)$ is:

(A) 64 (B) 32 (C) 51 (D) 48

Open solution

Q33. JEE Main 2025 (Online) 22nd January Morning Shift

Two balls are selected at random one by one without replacement from a bag containing 4 white and 6 black balls. If the probability that the first selected ball is black, given that the second selected ball is also black, is $\frac{m}{n}$, where $\gcd(m, n) = 1$, then $m + n$ is equal to :

(A) 4 (B) 14 (C) 11 (D) 13

Open solution

Q34. JEE Main 2024 (Online) 9th April Evening Shift

If an unbiased dice is rolled thrice, then the probability of getting a greater number in the ith roll than the number obtained in the $(i-1)^{\text{th}}$ roll, $i = 2, 3$, is equal to

(A) 5/54 (B) 2/54 (C) 1/54 (D) 3/54

Open solution

Q35. JEE Main 2024 (Online) 8th April Evening Shift

There are three bags X, Y and Z. Bag X contains 5 one-rupee coins and 4 five-rupee coins; Bag Y contains 4 one-rupee coins and 5 five-rupee coins and Bag Z contains 3 one-rupee coins and 6 five-rupee coins. A bag is selected at random and a coin drawn from it at random is found to be a one-rupee coin. Then the probability, that it came from bag Y, is :

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{5}{12}$ (D) $\frac{1}{4}$

Open solution

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Q36. JEE Main 2024 (Online) 8th April Morning Shift

Let the sum of two positive integers be 24 . If the probability, that their product is not less than $\frac{3}{4}$ times their greatest possible product, is $\frac{m}{n}$, where $\gcd(m, n) = 1$, then $n - m$ equals

(A) 10 (B) 11 (C) 9 (D) 8

Open solution

Q37. JEE Main 2024 (Online) 6th April Evening Shift

If three letters can be posted to any one of the 5 different addresses, then the probability that the three letters are posted to exactly two addresses is :

(A) $\frac{18}{25}$ (B) $\frac{12}{25}$ (C) $\frac{6}{25}$ (D) $\frac{4}{25}$

Open solution

Q38. JEE Main 2024 (Online) 6th April Morning Shift

A company has two plants A and B to manufacture motorcycles. 60% motorcycles are manufactured at plant A and the remaining are manufactured at plant B. 80% of the motorcycles manufactured at plant A are rated of the standard quality, while 90% of the motorcycles manufactured at plant B are rated of the standard quality. A motorcycle picked up randomly from the total production is found to be of the standard quality. If p is the probability that it was manufactured at plant B, then $126p$ is

(A) 54 (B) 66 (C) 56 (D) 64

Open solution

Q39. JEE Main 2024 (Online) 5th April Evening Shift

The coefficients $\mathrm{a}, \mathrm{b}, \mathrm{c}$ in the quadratic equation $\mathrm{ax}^2 + \mathrm{bx} + \mathrm{c} = 0$ are from the set $\{1, 2, 3, 4, 5, 6\}$. If the probability of this equation having one real root bigger than the other is p, then 216p equals :

(A) 38 (B) 76 (C) 57 (D) 19

Open solution

Q40. JEE Main 2024 (Online) 5th April Morning Shift

The coefficients a, b, c in the quadratic equation $ax^2 + bx + c = 0$ are chosen from the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$. The probability of this equation having repeated roots is :

(A) $\frac{1}{128}$ (B) $\frac{1}{64}$ (C) $\frac{3}{256}$ (D) $\frac{3}{128}$

Open solution

Q41. JEE Main 2024 (Online) 4th April Evening Shift

If the mean of the following probability distribution of a random variable X :

X	0	2	4	6	8
P(X)	a	$2a$	$a + b$	$2b$	$3b$

is $\frac{46}{9}$, then the variance of the distribution is

(A) $\frac{581}{81}$ (B) $\frac{566}{81}$ (C) $\frac{151}{27}$ (D) $\frac{173}{27}$

Open solution

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Q42. JEE Main 2024 (Online) 4th April Morning Shift

Three urns A, B and C contain 7 red, 5 black; 5 red, 7 black and 6 red, 6 black balls, respectively. One of the urn is selected at random and a ball is drawn from it. If the ball drawn is black, then the probability that it is drawn from urn A is :

Open solution

(A) $\frac{4}{17}$ (B) $\frac{5}{16}$ (C) $\frac{5}{18}$ (D) $\frac{7}{18}$

Q43. JEE Main 2024 (Online) 1st February Evening Shift

Let Ajay will not appear in JEE exam with probability $\mathrm{p}=\frac{2}{7}$, while both Ajay and Vijay will appear in the exam with probability $\mathrm{q}=\frac{1}{5}$. Then the probability, that Ajay will appear in the exam and Vijay will not appear is :

Open solution

(A) $\frac{9}{35}$ (B) $\frac{3}{35}$ (C) $\frac{24}{35}$ (D) $\frac{18}{35}$

Q44. JEE Main 2024 (Online) 1st February Morning Shift

A bag contains 8 balls, whose colours are either white or black. 4 balls are drawn at random without replacement and it was found that 2 balls are white and other 2 balls are black. The probability that the bag contains equal number of white and black balls is :

Open solution

(A) $\frac{2}{5}$ (B) $\frac{2}{7}$ (C) $\frac{1}{7}$ (D) $\frac{1}{5}$

Q45. JEE Main 2024 (Online) 31st January Evening Shift

A coin is biased so that a head is twice as likely to occur as a tail. If the coin is tossed 3 times, then the probability of getting two tails and one head is

Open solution

(A) $\frac{1}{9}$ (B) $\frac{2}{9}$ (C) $\frac{1}{27}$ (D) $\frac{2}{27}$

Q46. JEE Main 2024 (Online) 31st January Morning Shift

Three rotten apples are accidently mixed with fifteen good apples. Assuming the random variable x to be the number of rotten apples in a draw of two apples, the variance of x is

Open solution

(A) $\frac{57}{153}$ (B) $\frac{40}{153}$ (C) $\frac{37}{153}$ (D) $\frac{47}{153}$

Q47. JEE Main 2024 (Online) 31st January Morning Shift

Two marbles are drawn in succession from a box containing 10 red, 30 white, 20 blue and 15 orange marbles, with replacement being made after each drawing. Then the probability, that first drawn marble is red and second drawn marble is white, is

Open solution

(A) $\frac{4}{25}$ (B) $\frac{2}{3}$ (C) $\frac{2}{25}$ (D) $\frac{4}{75}$

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Q48. JEE Main 2024 (Online) 30th January Evening Shift

Bag A contains 3 white, 7 red balls and Bag B contains 3 white, 2 red balls. One bag is selected at random and a ball is drawn from it. The probability of drawing the ball from the bag A, if the ball drawn is white, is

(A) 1/4 (B) 1/3 (C) 3/10 (D) 1/9

Open solution

Q49. JEE Main 2024 (Online) 30th January Morning Shift

Two integers x and y are chosen with replacement from the set $\{0, 1, 2, 3, \ldots, 10\}$. Then the probability that $|x - y| > 5$, is :

(A) $\frac{31}{121}$ (B) $\frac{60}{121}$ (C) $\frac{62}{121}$ (D) $\frac{30}{121}$

Open solution

Q50. JEE Main 2024 (Online) 29th January Evening Shift

An integer is chosen at random from the integers $1, 2, 3, \ldots, 50$. The probability that the chosen integer is a multiple of atleast one of 4, 6 and 7 is

(A) $\frac{8}{25}$ (B) $\frac{9}{50}$ (C) $\frac{14}{25}$ (D) $\frac{21}{50}$

Open solution

Q51. JEE Main 2024 (Online) 29th January Morning Shift

A fair die is thrown until 2 appears. Then the probability, that 2 appears in even number of throws, is

(A) $\frac{5}{11}$ (B) $\frac{5}{6}$ (C) $\frac{1}{6}$ (D) $\frac{6}{11}$

Open solution

Q52. JEE Main 2024 (Online) 27th January Evening Shift

An urn contains 6 white and 9 black balls. Two successive draws of 4 balls are made without replacement. The probability, that the first draw gives all white balls and the second draw gives all black balls, is :

(A) $\frac{3}{256}$ (B) $\frac{5}{256}$ (C) $\frac{3}{715}$ (D) $\frac{5}{715}$

Open solution

Q53. JEE Main 2023 (Online) 15th April Morning Shift

A bag contains 6 white and 4 black balls. A die is rolled once and the number of balls equal to the number obtained on the die are drawn from the bag at random. The probability that all the balls drawn are white is :

(A) $\frac{1}{4}$ (B) $\frac{9}{50}$ (C) $\frac{1}{5}$ (D) $\frac{11}{50}$

Open solution

Q54. JEE Main 2023 (Online) 13th April Morning Shift

A coin is biased so that the head is 3 times as likely to occur as tail. This coin is tossed until a head or three tails occur. If X denotes the number of tosses of the coin, then the mean of X is :

(A) $\frac{81}{64}$ (B) $\frac{37}{16}$ (C) $\frac{21}{16}$ (D) $\frac{15}{16}$

Open solution

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Q55. JEE Main 2023 (Online) 12th April Morning Shift

Two dice A and B are rolled. Let the numbers obtained on A and B be α and β respectively. If the variance of $\alpha - \beta$ is $\frac{p}{q}$, where p and q are co-prime, then the sum of the positive divisors of p is equal to :

Open solution

(A) 48 (B) 31 (C) 72 (D) 36

Q56. JEE Main 2023 (Online) 11th April Morning Shift

Let $S = \{M = [a_{ij}], a_{ij} \in \{0, 1, 2\}, 1 \le i, j \le 2\}$ be a sample space and $A = \{M \in S : M$ is invertible $\}$ be an event. Then $P(A)$ is equal to :

Open solution

(A) $\frac{47}{81}$ (B) $\frac{49}{81}$ (C) $\frac{50}{81}$ (D) $\frac{16}{27}$

Q57. JEE Main 2023 (Online) 10th April Morning Shift

Let N denote the sum of the numbers obtained when two dice are rolled. If the probability that $2^N < N!$ is $\frac{m}{n}$, where m and n are coprime, then $4m - 3n$ is equal to :

Open solution

(A) 12 (B) 6 (C) 8 (D) 10

Q58. JEE Main 2023 (Online) 8th April Evening Shift

If the probability that the random variable X takes values x is given by $\mathrm{P}(\mathrm{X} = x) = \mathrm{k}(x + 1)3^{-x}, x = 0, 1, 2, 3, \ldots$, where k is a constant, then $\mathrm{P}(\mathrm{X} \ge 2)$ is equal to :

Open solution

(A) $\frac{7}{18}$ (B) $\frac{20}{27}$ (C) $\frac{7}{27}$ (D) $\frac{11}{18}$

Q59. JEE Main 2023 (Online) 8th April Morning Shift

In a bolt factory, machines A, B and C manufacture respectively $20\%, 30\%$ and 50% of the total bolts. Of their output 3, 4 and 2 percent are respectively defective bolts. A bolt is drawn at random from the product. If the bolt drawn is found the defective, then the probability that it is manufactured by the machine C is :

Open solution

(A) $\frac{2}{7}$ (B) $\frac{9}{28}$ (C) $\frac{5}{14}$ (D) $\frac{3}{7}$

Q60. JEE Main 2023 (Online) 6th April Evening Shift

Three dice are rolled. If the probability of getting different numbers on the three dice is $\frac{p}{q}$, where p and q are co-prime, then $q - p$ is equal to :

Open solution

(A) 3 (B) 4 (C) 1 (D) 2

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Q61. JEE Main 2023 (Online) 1st February Evening Shift

Two dice are thrown independently. Let A be the event that the number appeared on the 1st die is less than the number appeared on the 2nd die, B be the event that the number appeared on the 1st die is even and that on the second die is odd, and C be the event that the number appeared on the 1st die is odd and that on the 2nd is even. Then :

(A) A and B are mutually exclusive

(B) the number of favourable cases of the events A, B and C are 15, 6 and 6 respectively

(C) B and C are independent

(D) the number of favourable cases of the event $(\mathrm{A} \cup \mathrm{B}) \cap \mathrm{C}$ is 6

Open solution

Q62. JEE Main 2023 (Online) 31st January Morning Shift

A bag contains 6 balls. Two balls are drawn from it at random and both are found to be black. The probability that the bag contains at least 5 black balls is :

(A) $\frac{3}{7}$ (B) $\frac{5}{6}$ (C) $\frac{5}{7}$ (D) $\frac{2}{7}$

Open solution

Q63. JEE Main 2023 (Online) 30th January Morning Shift

If an unbiased die, marked with $-2, -1, 0, 1, 2, 3$ on its faces, is thrown five times, then the probability that the product of the outcomes is positive, is :

(A) $\frac{27}{288}$ (B) $\frac{521}{2592}$ (C) $\frac{440}{2592}$ (D) $\frac{881}{2592}$

Open solution

Q64. JEE Main 2023 (Online) 29th January Evening Shift

Let $\mathrm{S} = \{w_1, w_2, \ldots\ldots\}$ be the sample space associated to a random experiment. Let $P(w_n) = \frac{P(w_{n-1})}{2}, n \geq 2$. Let $A = \{2k + 3l : k, l \in N\}$ and $B = \{w_n : n \in A\}$. Then P(B) is equal to :

(A) $\frac{3}{32}$ (B) $\frac{1}{32}$ (C) $\frac{1}{16}$ (D) $\frac{3}{64}$

Open solution

Q65. JEE Main 2023 (Online) 29th January Morning Shift

Three rotten apples are mixed accidently with seven good apples and four apples are drawn one by one without replacement. Let the random variable X denote the number of rotten apples. If μ and σ^2 represent mean and variance of X, respectively, then $10(\mu^2 + \sigma^2)$ is equal to :

(A) 20 (B) 30 (C) 250 (D) 25

Open solution

Q66. JEE Main 2023 (Online) 29th January Morning Shift

Fifteen football players of a club-team are given 15 T-shirts with their names written on the backside. If the players pick up the T-shirts randomly, then the probability that at least 3 players pick the correct T-shirt is :

(A) $\frac{1}{6}$ (B) $\frac{2}{15}$ (C) $\frac{5}{24}$ (D) 0.08

Open solution

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Q67. JEE Main 2023 (Online) 25th January Evening Shift

Let N be the sum of the numbers appeared when two fair dice are rolled and let the probability that $N-2, \sqrt{3N}, N+2$ are in geometric progression be $\frac{k}{48}$. Then the value of k is :

(A) 8 (B) 16 (C) 2 (D) 4

Open solution

Q68. JEE Main 2023 (Online) 25th January Morning Shift

Let M be the maximum value of the product of two positive integers when their sum is 66. Let the sample space $S = \{x \in \mathbb{Z} : x(66-x) \geq \frac{5}{9}M\}$ and the event $\mathrm{A} = \{\mathrm{x} \in \mathrm{S} : \mathrm{x\ is\ a\ multiple\ of\ 3}\}$. Then P(A) is equal to :

(A) $\frac{1}{3}$ (B) $\frac{1}{5}$ (C) $\frac{7}{22}$ (D) $\frac{15}{44}$

Open solution

Q69. JEE Main 2023 (Online) 24th January Morning Shift

Let N denote the number that turns up when a fair die is rolled. If the probability that the system of equations

$x + y + z = 1$

$2x + \mathrm{N}y + 2z = 2$

$3x + 3y + \mathrm{N}z = 3$

has unique solution is $\frac{k}{6}$, then the sum of value of k and all possible values of N is :

(A) 18 (B) 21 (C) 20 (D) 19

Open solution

Q70. JEE Main 2023 (Online) 24th January Morning Shift

Let Ω be the sample space and $\mathrm{A} \subseteq \Omega$ be an event.

Given below are two statements :

(S1) : If P(A) = 0, then A = ϕ

(S2) : If P(A) = 1, then A = Ω

Then :

(A) both (S1) and (S2) are true (B) both (S1) and (S2) are false

(C) only (S2) is true (D) only (S1) is true

Open solution

Q71. JEE Main 2022 (Online) 29th July Evening Shift

Bag I contains 3 red, 4 black and 3 white balls and Bag II contains 2 red, 5 black and 2 white balls. One ball is transferred from Bag I to Bag II and then a ball is drawn from Bag II. The ball so drawn is found to be black in colour. Then the probability, that the transferred ball is red, is :

(A) $\frac{4}{9}$ (B) $\frac{5}{18}$ (C) $\frac{1}{6}$ (D) $\frac{3}{10}$

Open solution

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Q72. JEE Main 2022 (Online) 29th July Morning Shift

Let $S = \{1, 2, 3, \ldots, 2022\}$. Then the probability, that a randomly chosen number n from the set S such that $\mathrm{HCF}\,(\mathrm{n}, 2022) = 1$, is :

(A) $\frac{128}{1011}$ (B) $\frac{166}{1011}$ (C) $\frac{127}{337}$ (D) $\frac{112}{337}$

Open solution

Q73. JEE Main 2022 (Online) 28th July Evening Shift

Let A and B be two events such that $P(B \mid A) = \frac{2}{5}, P(A \mid B) = \frac{1}{7}$ and $P(A \cap B) = \frac{1}{9}$. Consider

(S1) $P\left(A' \cup B\right) = \frac{5}{6}$,

(S2) $P\left(A' \cap B'\right) = \frac{1}{18}$

Then :

(A) Both (S1) and (S2) are true

(B) Both (S1) and (S2) are false

(C) Only (S1) is true

(D) Only (S2) is true

Open solution

Q74. JEE Main 2022 (Online) 28th July Morning Shift

Out of 60% female and 40% male candidates appearing in an exam, 60% candidates qualify it. The number of females qualifying the exam is twice the number of males qualifying it. A candidate is randomly chosen from the qualified candidates. The probability, that the chosen candidate is a female, is :

(A) $\frac{2}{3}$ (B) $\frac{11}{16}$ (C) $\frac{23}{32}$ (D) $\frac{13}{16}$

Open solution

Q75. JEE Main 2022 (Online) 27th July Evening Shift

A six faced die is biased such that

$3 \times \mathrm{P}(\text{a prime number}) = 6 \times \mathrm{P}(\text{a composite number}) = 2 \times \mathrm{P}(1)$.

Let X be a random variable that counts the number of times one gets a perfect square on some throws of this die. If the die is thrown twice, then the mean of X is :

(A) $\frac{3}{11}$ (B) $\frac{5}{11}$ (C) $\frac{7}{11}$ (D) $\frac{8}{11}$

Open solution

Q76. JEE Main 2022 (Online) 27th July Morning Shift

Let S be the sample space of all five digit numbers. If p is the probability that a randomly selected number from S, is a multiple of 7 but not divisible by 5 , then $9p$ is equal to :

(A) 1.0146 (B) 1.2085 (C) 1.0285 (D) 1.1521

Open solution

Q77. JEE Main 2022 (Online) 26th July Morning Shift

Let $\mathrm{E}_1, \mathrm{E}_2, \mathrm{E}_3$ be three mutually exclusive events such that $\mathrm{P}\left(\mathrm{E}_1\right) = \frac{2+3\mathrm{p}}{6}, \mathrm{P}\left(\mathrm{E}_2\right) = \frac{2-\mathrm{p}}{8}$ and $\mathrm{P}\left(\mathrm{E}_3\right) = \frac{1-\mathrm{p}}{2}$. If the maximum and minimum values of p are p_1 and p_2, then $\left(\mathrm{p}_1 + \mathrm{p}_2\right)$ is equal to :

(A) $\frac{2}{3}$ (B) $\frac{5}{3}$ (C) $\frac{5}{4}$ (D) 1

Open solution

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Q78. JEE Main 2022 (Online) 25th July Evening Shift

If A and B are two events such that $P(A) = \frac{1}{3}, P(B) = \frac{1}{5}$ and $P(A \cup B) = \frac{1}{2}$, then $P(A \mid B') + P(B \mid A')$ is equal to :

(A) $\frac{3}{4}$ (B) $\frac{5}{8}$ (C) $\frac{5}{4}$ (D) $\frac{7}{8}$

Open solution

Q79. JEE Main 2022 (Online) 25th July Morning Shift

If the numbers appeared on the two throws of a fair six faced die are α and β, then the probability that $x^2 + \alpha x + \beta > 0$, for all $x \in \mathbf{R}$, is :

(A) $\frac{17}{36}$ (B) $\frac{4}{9}$ (C) $\frac{1}{2}$ (D) $\frac{19}{36}$

Open solution

Q80. JEE Main 2022 (Online) 29th June Morning Shift

The probability that a randomly chosen 2 $\times$ 2 matrix with all the entries from the set of first 10 primes, is singular, is equal to :

(A) $\frac{133}{10^4}$ (B) $\frac{18}{10^3}$ (C) $\frac{19}{10^3}$ (D) $\frac{271}{10^4}$

Open solution

Q81. JEE Main 2022 (Online) 28th June Evening Shift

The probability that a randomly chosen one-one function from the set {a, b, c, d} to the set {1, 2, 3, 4, 5} satisfies f(a) + 2f(b) $-$ f(c) = f(d) is :

(A) $\frac{1}{24}$ (B) $\frac{1}{40}$ (C) $\frac{1}{30}$ (D) $\frac{1}{20}$

Open solution

Q82. JEE Main 2022 (Online) 28th June Morning Shift

The probability, that in a randomly selected 3-digit number at least two digits are odd, is :

(A) $\frac{19}{36}$ (B) $\frac{15}{36}$ (C) $\frac{13}{36}$ (D) $\frac{23}{36}$

Open solution

Q83. JEE Main 2022 (Online) 27th June Evening Shift

If a point A(x, y) lies in the region bounded by the y-axis, straight lines 2y + x = 6 and 5x $-$ 6y = 30, then the probability that y < 1 is :

(A) $\frac{1}{6}$ (B) $\frac{5}{6}$ (C) $\frac{2}{3}$ (D) $\frac{6}{7}$

Open solution

Q84. JEE Main 2022 (Online) 27th June Morning Shift

Five numbers x_1, x_2, x_3, x_4, x_5 are randomly selected from the numbers 1, 2, 3, ..., 18 and are arranged in the increasing order $(x_1 < x_2 < x_3 < x_4 < x_5)$. The probability that $x_2 = 7$ and $x_4 = 11$ is :

(A) $\frac{1}{136}$ (B) $\frac{1}{72}$ (C) $\frac{1}{68}$ (D) $\frac{1}{34}$

Open solution

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Q85. JEE Main 2022 (Online) 25th June Evening Shift

A biased die is marked with numbers 2, 4, 8, 16, 32, 32 on its faces and the probability of getting a face with mark n is $\frac{1}{n}$. If the die is thrown thrice, then the probability, that the sum of the numbers obtained is 48, is :

Open solution

(A) $\frac{7}{2^{11}}$ (B) $\frac{7}{2^{12}}$ (C) $\frac{3}{2^{10}}$ (D) $\frac{13}{2^{12}}$

Q86. JEE Main 2022 (Online) 25th June Morning Shift

Let E_1 and E_2 be two events such that the conditional probabilities $P(E_1|E_2) = \frac{1}{2}$, $P(E_2|E_1) = \frac{3}{4}$ and $P(E_1 \cap E_2) = \frac{1}{8}$. Then :

Open solution

(A) $P(E_1 \cap E_2) = P(E_1) \,.\, P(E_2)$ (B) $P(E_1' \cap E_2') = P(E_1') \,.\, P(E_2)$

(C) $P(E_1 \cap E_2') = P(E_1) \,.\, P(E_2)$ (D) $P(E_1' \cap E_2) = P(E_1) \,.\, P(E_2)$

Q87. JEE Main 2022 (Online) 24th June Evening Shift

A random variable X has the following probability distribution :

Open solution

X	0	1	2	3	4
P(X)	k	2k	4k	6k	8k

The value of P(1 < X < 4 | X $\leq$ 2) is equal to :

(A) $\frac{4}{7}$ (B) $\frac{2}{3}$ (C) $\frac{3}{7}$ (D) $\frac{4}{5}$

Q88. JEE Main 2022 (Online) 24th June Morning Shift

Bag A contains 2 white, 1 black and 3 red balls and bag B contains 3 black, 2 red and n white balls. One bag is chosen at random and 2 balls drawn from it at random, are found to be 1 red and 1 black. If the probability that both balls come from Bag A is $\frac{6}{11}$, then n is equal to _____.

Open solution

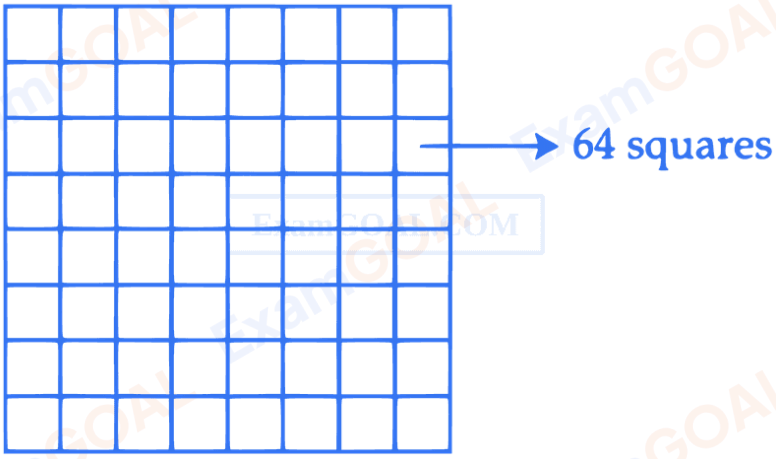
(A) 13 (B) 6 (C) 4 (D) 3

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Q89. JEE Main 2021 (Online) 1st September Evening Shift

Two squares are chosen at random on a chessboard (see figure). The probability that they have a side in common is :

Open solution



(A) $\frac{2}{7}$ (B) $\frac{1}{18}$ (C) $\frac{1}{7}$ (D) $\frac{1}{9}$

Q90. JEE Main 2021 (Online) 31st August Evening Shift

Let S = {1, 2, 3, 4, 5, 6}. Then the probability that a randomly chosen onto function g from S to S satisfies g(3) = 2g(1) is :

Open solution

(A) $\frac{1}{10}$ (B) $\frac{1}{15}$ (C) $\frac{1}{5}$ (D) $\frac{1}{30}$

Q91. JEE Main 2021 (Online) 27th August Morning Shift

When a certain biased die is rolled, a particular face occurs with probability $\frac{1}{6} - x$ and its opposite face occurs with probability $\frac{1}{6} + x$. All other faces occur with probability $\frac{1}{6}$. Note that opposite faces sum to 7 in any die. If $0 < x < \frac{1}{6}$, and the probability of obtaining total sum = 7, when such a die is rolled twice, is $\frac{13}{96}$, then the value of x is :

Open solution

(A) $\frac{1}{16}$ (B) $\frac{1}{8}$ (C) $\frac{1}{9}$ (D) $\frac{1}{12}$

Q92. JEE Main 2021 (Online) 26th August Evening Shift

A fair die is tossed until six is obtained on it. Let x be the number of required tosses, then the conditional probability $P(x \geq 5 \mid x > 2)$ is :

Open solution

(A) $\frac{125}{216}$ (B) $\frac{11}{36}$ (C) $\frac{5}{6}$ (D) $\frac{25}{36}$

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Q93. JEE Main 2021 (Online) 26th August Evening Shift

Open solution

Two fair dice are thrown. The numbers on them are taken as λ and μ, and a system of linear equations

x + y + z = 5

x + 2y + 3z = μ

x + 3y + λz = 1

is constructed. If p is the probability that the system has a unique solution and q is the probability that the system has no solution, then :

(A) $p = \frac{1}{6}$ and $q = \frac{1}{36}$

(B) $p = \frac{5}{6}$ and $q = \frac{5}{36}$

(C) $p = \frac{5}{6}$ and $q = \frac{1}{36}$

(D) $p = \frac{1}{6}$ and $q = \frac{5}{36}$

Q94. JEE Main 2021 (Online) 26th August Morning Shift

Open solution

Let A and B be independent events such that P(A) = p, P(B) = 2p. The largest value of p, for which P (exactly one of A, B occurs) = $\frac{5}{9}$, is :

(A) $\frac{1}{3}$ (B) $\frac{2}{9}$ (C) $\frac{4}{9}$ (D) $\frac{5}{12}$

Q95. JEE Main 2021 (Online) 27th July Morning Shift

Open solution

The probability that a randomly selected 2-digit number belongs to the set {n $\in$ N : ($2^n - 2$) is a multiple of 3} is equal to :

(A) $\frac{1}{6}$ (B) $\frac{2}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{3}$

Q96. JEE Main 2021 (Online) 25th July Evening Shift

Open solution

Let X be a random variable such that the probability function of a distribution is given by $P(X = 0) = \frac{1}{2}, P(X = j) = \frac{1}{3^j}(j = 1, 2, 3, \ldots, \infty)$. Then the mean of the distribution and P(X is positive and even) respectively are :

(A) $\frac{3}{8}$ and $\frac{1}{8}$

(B) $\frac{3}{4}$ and $\frac{1}{8}$

(C) $\frac{3}{4}$ and $\frac{1}{9}$

(D) $\frac{3}{4}$ and $\frac{1}{16}$

Q97. JEE Main 2021 (Online) 25th July Morning Shift

Open solution

Let 9 distinct balls be distributed among 4 boxes, B_1, B_2, B_3 and B_4. If the probability that B_3 contains exactly 3 balls is $k\left(\frac{3}{4}\right)^9$ then k lies in the set :

(A) {x $\in$ R : |x $-$ 3| < 1}

(B) {x $\in$ R : |x $-$ 2| $\leq$ 1}

(C) {x $\in$ R : |x $-$ 1| < 1}

(D) {x $\in$ R : |x $-$ 5| $\leq$ 1}

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Q98. JEE Main 2021 (Online) 22th July Evening Shift

Four dice are thrown simultaneously and the numbers shown on these dice are recorded in 2×2 matrices. The probability that such formed matrix have all different entries and are non-singular, is :

(A) $\frac{45}{162}$ (B) $\frac{21}{81}$ (C) $\frac{22}{81}$ (D) $\frac{43}{162}$

Open solution

Q99. JEE Main 2021 (Online) 20th July Evening Shift

Let A, B and C be three events such that the probability that exactly one of A and B occurs is $(1 - k)$, the probability that exactly one of B and C occurs is $(1 - 2k)$, the probability that exactly one of C and A occurs is $(1 - k)$ and the probability of all A, B and C occur simultaneously is k^2, where $0 < k < 1$. Then the probability that at least one of A, B and C occur is :

(A) greater than $\frac{1}{8}$ but less than $\frac{1}{4}$ (B) greater than $\frac{1}{2}$

(C) greater than $\frac{1}{4}$ but less than $\frac{1}{2}$ (D) exactly equal to $\frac{1}{2}$

Open solution

Q100. JEE Main 2021 (Online) 20th July Morning Shift

Words with or without meaning are to be formed using all the letters of the word EXAMINATION. The probability that the letter M appears at the fourth position in any such word is :

(A) $\frac{1}{66}$ (B) $\frac{1}{11}$ (C) $\frac{1}{9}$ (D) $\frac{2}{11}$

Open solution

Q101. JEE Main 2021 (Online) 20th July Morning Shift

The probability of selecting integers $a \in [-5, 30]$ such that $x^2 + 2(a + 4)x - 5a + 64 > 0$, for all $x \in R$, is :

(A) $\frac{7}{36}$ (B) $\frac{2}{9}$ (C) $\frac{1}{6}$ (D) $\frac{1}{4}$

Open solution

Q102. JEE Main 2021 (Online) 17th March Evening Shift

Let a computer program generate only the digits 0 and 1 to form a string of binary numbers with probability of occurrence of 0 at even places be $\frac{1}{2}$ and probability of occurrence of 0 at the odd place be $\frac{1}{3}$. Then the probability that '10' is followed by '01' is equal to :

(A) $\frac{1}{18}$ (B) $\frac{1}{3}$ (C) $\frac{1}{9}$ (D) $\frac{1}{6}$

Open solution

Q103. JEE Main 2021 (Online) 17th March Morning Shift

Two dies are rolled. If both dices have six faces numbered 1, 2, 3, 5, 7 and 11, then the probability that the sum of the numbers on the top faces is less than or equal to 8 is :

(A) $\frac{4}{9}$ (B) $\frac{1}{2}$ (C) $\frac{5}{12}$ (D) $\frac{17}{36}$

Open solution

Q104. JEE Main 2021 (Online) 16th March Evening Shift

Let A denote the event that a 6-digit integer formed by 0, 1, 2, 3, 4, 5, 6 without repetitions, be divisible by 3. Then probability of event A is equal to :

(A) $\frac{4}{9}$ (B) $\frac{9}{56}$ (C) $\frac{11}{27}$ (D) $\frac{3}{7}$

Open solution

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Q105. JEE Main 2021 (Online) 16th March Morning Shift

A pack of cards has one card missing. Two cards are drawn randomly and are found to be spades. The probability that the missing card is not a spade, is :

(A) $\frac{39}{50}$ (B) $\frac{3}{4}$ (C) $\frac{22}{425}$ (D) $\frac{52}{867}$

Open solution

Q106. JEE Main 2021 (Online) 26th February Evening Shift

A seven digit number is formed using digits 3, 3, 4, 4, 4, 5, 5. The probability, that number so formed is divisible by 2, is :

(A) $\frac{1}{7}$ (B) $\frac{4}{7}$ (C) $\frac{6}{7}$ (D) $\frac{3}{7}$

Open solution

Q107. JEE Main 2021 (Online) 25th February Evening Shift

Let A be a set of all 4-digit natural numbers whose exactly one digit is 7. Then the probability that a randomly chosen element of A leaves remainder 2 when divided by 5 is :

(A) $\frac{2}{9}$ (B) $\frac{1}{5}$ (C) $\frac{122}{297}$ (D) $\frac{97}{297}$

Open solution

Q108. JEE Main 2021 (Online) 25th February Evening Shift

In a group of 400 people, 160 are smokers and non-vegetarian; 100 are smokers and vegetarian and the remaining 140 are non-smokers and vegetarian. Their chances of getting a particular chest disorder are 35%, 20% and 10% respectively. A person is chosen from the group at random and is found to be suffering from the chest disorder. The probability that the selected person is a smoker and non-vegetarian is :

(A) $\frac{14}{45}$ (B) $\frac{8}{45}$ (C) $\frac{7}{45}$ (D) $\frac{28}{45}$

Open solution

Q109. JEE Main 2021 (Online) 25th February Morning Shift

When a missile is fired from a ship, the probability that it is intercepted is $\frac{1}{3}$ and the probability that the missile hits the target, given that it is not intercepted, is $\frac{3}{4}$. If three missiles are fired independently from the ship, then the probability that all three hit the target, is :

(A) $\frac{3}{4}$ (B) $\frac{3}{8}$ (C) $\frac{1}{27}$ (D) $\frac{1}{8}$

Open solution

Q110. JEE Main 2021 (Online) 25th February Morning Shift

The coefficients a, b and c of the quadratic equation, $ax^2 + bx + c = 0$ are obtained by throwing a dice three times. The probability that this equation has equal roots is :

(A) $\frac{1}{72}$ (B) $\frac{5}{216}$ (C) $\frac{1}{36}$ (D) $\frac{1}{54}$

Open solution

Q111. JEE Main 2021 (Online) 24th February Evening Shift

The probability that two randomly selected subsets of the set {1, 2, 3, 4, 5} have exactly two elements in their intersection, is :

(A) $\frac{135}{2^9}$ (B) $\frac{65}{2^8}$ (C) $\frac{65}{2^7}$ (D) $\frac{35}{2^7}$

Open solution

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Q112. JEE Main 2020 (Online) 6th September Evening Slot

The probabilities of three events A, B and C are given by
P(A) = 0.6, P(B) = 0.4 and P(C) = 0.5.
If $P(A \cup B) = 0.8$, $P(A \cap C) = 0.3$, $P(A \cap B \cap C) = 0.2$, $P(B \cap C) = \beta$
and $P(A \cup B \cup C) = \alpha$, where $0.85 \leq \alpha \leq 0.95$, then β lies in the interval :

(A) [0.35, 0.36] (B) [0.20, 0.25] (C) [0.25, 0.35] (D) [0.36, 0.40]

Open solution

Q113. JEE Main 2020 (Online) 6th September Morning Slot

Out of 11 consecutive natural numbers if three numbers are selected at random (without repetition), then the probability that they are in A.P. with positive common difference, is :

(A) $\frac{10}{99}$ (B) $\frac{5}{33}$ (C) $\frac{15}{101}$ (D) $\frac{5}{101}$

Open solution

Q114. JEE Main 2020 (Online) 4th September Evening Slot

In a game two players A and B take turns in throwing a pair of fair dice starting with player A and total of scores on the two dice, in each throw is noted. A wins the game if he throws a total of 6 before B throws a total of 7 and B wins the game if he throws a total of 7 before A throws a total of six. The game stops as soon as either of the players wins. The probability of A winning the game is :

(A) $\frac{5}{6}$ (B) $\frac{5}{31}$ (C) $\frac{31}{61}$ (D) $\frac{30}{61}$

Open solution

Q115. JEE Main 2020 (Online) 3rd September Evening Slot

The probability that a randomly chosen 5-digit number is made from exactly two digits is :

(A) $\frac{150}{10^4}$ (B) $\frac{134}{10^4}$ (C) $\frac{121}{10^4}$ (D) $\frac{135}{10^4}$

Open solution

Q116. JEE Main 2020 (Online) 3rd September Morning Slot

A dice is thrown two times and the sum of the scores appearing on the die is observed to be a multiple of 4. Then the conditional probability that the score 4 has appeared atleast once is :

(A) $\frac{1}{8}$ (B) $\frac{1}{9}$ (C) $\frac{1}{4}$ (D) $\frac{1}{3}$

Open solution

Q117. JEE Main 2020 (Online) 2nd September Evening Slot

Let E^C denote the complement of an event E. Let E_1, E_2 and E_3 be any pairwise independent events with $P(E_1) > 0$

and $P(E_1 \cap E_2 \cap E_3) = 0$.

Then $P(E_2^C \cap E_3^C / E_1)$ is equal to :

(A) $P(E_3^C) - P(E_2)$ (B) $P(E_2^C) + P(E_3)$

(C) $P(E_3^C) - P(E_2^C)$ (D) $P(E_3) - P(E_2^C)$

Open solution

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Q118. JEE Main 2020 (Online) 2nd September Morning Slot

Box I contains 30 cards numbered 1 to 30 and Box II contains 20 cards numbered 31 to 50. A box is selected at random and a card is drawn from it. The number on the card is found to be a non-prime number. The probability that the card was drawn from Box I is :

Open solution

(A) $\frac{8}{17}$ (B) $\frac{2}{3}$ (C) $\frac{2}{5}$ (D) $\frac{4}{17}$

Q119. JEE Main 2020 (Online) 9th January Evening Slot

A random variable X has the following probability distribution :

Open solution

X:	1	2	3	4	5
P(X):	K^2	2K	K	2K	$5K^2$

Then P(X > 2) is equal to :

(A) $\frac{1}{6}$ (B) $\frac{7}{12}$ (C) $\frac{1}{36}$ (D) $\frac{23}{36}$

Q120. JEE Main 2020 (Online) 9th January Evening Slot

If 10 different balls are to be placed in 4 distinct boxes at random, then the probability that two of these boxes contain exactly 2 and 3 balls is :

Open solution

(A) $\frac{965}{2^{11}}$ (B) $\frac{965}{2^{10}}$ (C) $\frac{945}{2^{11}}$ (D) $\frac{945}{2^{10}}$

Q121. JEE Main 2020 (Online) 9th January Morning Slot

In a box, there are 20 cards, out of which 10 are labelled as A and the remaining 10 are labelled as B. Cards are drawn at random, one after the other and with replacement, till a second A-card is obtained. The probability that the second A-card appears before the third B-card is :

Open solution

(A) $\frac{13}{16}$ (B) $\frac{11}{16}$ (C) $\frac{15}{16}$ (D) $\frac{9}{16}$

Q122. JEE Main 2020 (Online) 8th January Evening Slot

Let A and B be two events such that the probability that exactly one of them occurs is $\frac{2}{5}$ and the probability that A or B occurs is $\frac{1}{2}$, then the probability of both of them occur together is :

Open solution

(A) 0.20 (B) 0.02 (C) 0.01 (D) 0.10

Q123. JEE Main 2020 (Online) 8th January Morning Slot

Let A and B be two independent events such that
P(A) = $\frac{1}{3}$ and P(B) = $\frac{1}{6}$.
Then, which of the following is TRUE?

Open solution

(A) $P\left(\frac{A}{A \cup B}\right) = \frac{1}{4}$ (B) $P\left(\frac{A}{B}\right) = \frac{2}{3}$

(C) $P\left(\frac{A'}{B'}\right) = \frac{1}{3}$ (D) $P\left(\frac{A}{B'}\right) = \frac{1}{3}$

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Q124. JEE Main 2020 (Online) 7th January Morning Slot

An unbiased coin is tossed 5 times. Suppose that a variable X is assigned the value of k when k consecutive heads are obtained for k = 3, 4, 5, otherwise X takes the value -1. Then the expected value of X, is :

(A) $-\frac{3}{16}$ (B) $-\frac{1}{8}$ (C) $\frac{1}{8}$ (D) $\frac{3}{16}$

Open solution

Q125. JEE Main 2019 (Online) 12th April Evening Slot

A person throws two fair dice. He wins Rs. 15 for throwing a doublet (same numbers on the two dice), wins Rs. 12 when the throw results in the sum of 9, and loses Rs. 6 for any other outcome on the throw. Then the expected gain/loss (in Rs.) of the person is :

(A) $\frac{1}{4}$ loss (B) $\frac{1}{2}$ gain (C) $\frac{1}{2}$ loss (D) 2 gain

Open solution

Q126. JEE Main 2019 (Online) 12th April Morning Slot

If three of the six vertices of a regular hexagon are chosen at random, then the probability that the triangle formed with these chosen vertices is equilateral is :

(A) $\frac{1}{10}$ (B) $\frac{3}{10}$ (C) $\frac{3}{20}$ (D) $\frac{1}{5}$

Open solution

Q127. JEE Main 2019 (Online) 10th April Evening Slot

Minimum number of times a fair coin must be tossed so that the probability of getting at least one head is more than 99% is :

(A) 6 (B) 5 (C) 8 (D) 7

Open solution

Q128. JEE Main 2019 (Online) 10th April Morning Slot

Assume that each born child is equally likely to be a boy or a girl. If two families have two children each, then the conditional probability that all children are girls given that at least two are girls is :

(A) $\frac{1}{10}$ (B) $\frac{1}{17}$ (C) $\frac{1}{11}$ (D) $\frac{1}{12}$

Open solution

Q129. JEE Main 2019 (Online) 9th April Morning Slot

Four persons can hit a target correctly with probabilities $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$ and $\frac{1}{8}$ respectively. If all hit at the target independently, then the probability that the target would be hit, is :

(A) $\frac{25}{32}$ (B) $\frac{25}{192}$ (C) $\frac{1}{192}$ (D) $\frac{7}{32}$

Open solution

Q130. JEE Main 2019 (Online) 8th April Evening Slot

The minimum number of times one has to toss a fair coin so that the probability of observing at least one head is at least 90% is :

(A) 2 (B) 3 (C) 4 (D) 5

Open solution

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Q131. JEE Main 2019 (Online) 8th April Morning Slot

Let A and B be two non-null events such that $A \subset B$. Then, which of the following statements is always correct?

(A) $P(A|B) = 1$ (B) $P(A|B) = P(B) - P(A)$

(C) $P(A|B) \leq P(A)$ (D) $P(A|B) \geq P(A)$

Open solution

Q132. JEE Main 2019 (Online) 12th January Evening Slot

In a game, a man wins Rs. 100 if he gets 5 or 6 on a throw of a fair die and loses Rs. 50 for getting any other number on the die. If he decides to throw the die either till he gets a five or a six or to a maximum of three throws, then his expected gain/loss (in rupees) is :

(A) $\frac{400}{3}$ loss (B) 0 (C) $\frac{400}{9}$ loss (D) $\frac{400}{3}$ gain

Open solution

Q133. JEE Main 2019 (Online) 12th January Evening Slot

In a class of 60 students, 40 opted for NCC, 30 opted for NSS and 20 opted for both NCC and NSS. If one of these students is selected at random, then the probability that the students selected has opted neither for NCC nor for NSS is :

(A) $\frac{1}{3}$ (B) $\frac{1}{6}$ (C) $\frac{2}{3}$ (D) $\frac{5}{6}$

Open solution

Q134. JEE Main 2019 (Online) 12th January Morning Slot

In a random experiment, a fair die is rolled until two fours are obtained in succession. The probability that the experiment will end in the fifth throw of the die is equal to :

(A) $\frac{200}{6^5}$ (B) $\frac{225}{6^5}$ (C) $\frac{150}{6^5}$ (D) $\frac{175}{6^5}$

Open solution

Q135. JEE Main 2019 (Online) 11th January Evening Slot

Let S = {1, 2, ..., 20}. A subset B of S is said to be "nice", if the sum of the elements of B is 203. Then the probability that a randomly chosen subset of S is "nice" is :

(A) $\frac{5}{2^{20}}$ (B) $\frac{7}{2^{20}}$ (C) $\frac{4}{2^{20}}$ (D) $\frac{6}{2^{20}}$

Open solution

Q136. JEE Main 2019 (Online) 11th January Morning Slot

Two integers are selected at random from the set {1, 2, ..., 11}. Given that the sum of selected numbers is even, the conditional probability that both the numbers are even is :

(A) $\frac{2}{5}$ (B) $\frac{1}{2}$ (C) $\frac{7}{10}$ (D) $\frac{3}{5}$

Open solution

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Q137. JEE Main 2019 (Online) 10th January Morning Slot

An unbiased coin is tossed. If the outcome is a head then a pair of unbiased dice is rolled and the sum of the numbers obtained on them is noted. If the toss of the coin results in tail then a card from a well-shuffled pack of nine cards numbered 1, 2, 3, ..., 9 is randomly picked and the number on the card is noted. The probability that the noted number is either 7 or 8 is :

Open solution

(A) $\frac{19}{36}$ (B) $\frac{15}{72}$ (C) $\frac{13}{36}$ (D) $\frac{19}{72}$

Q138. JEE Main 2019 (Online) 9th January Evening Slot

An urn contains 5 red and 2 green balls. A ball is drawn at random from the urn. If the drawn ball is green, then a red ball is added to the urn and if the drawn ball is red, then a green ball is added to the urn; the original ball is not returned to the urn. Now, a second ball is drawn at random from it. The probability that the second ball is red, is :

Open solution

(A) $\frac{21}{49}$ (B) $\frac{27}{49}$ (C) $\frac{26}{49}$ (D) $\frac{32}{49}$

Q139. JEE Main 2018 (Online) 16th April Morning Slot

Let A, B and C be three events, which are pair-wise independent and $\overline{E}$ denotes the complement of an event E. If $P(A \cap B \cap C) = 0$ and $P(C) > 0$, then $P\left[\left(\overline{A} \cap \overline{B}\right)|C\right]$ is equal to :

Open solution

(A) $P\left(\overline{A}\right) - P(B)$ (B) $P(A) + P\left(\overline{B}\right)$

(C) $P\left(\overline{A}\right) - P\left(\overline{B}\right)$ (D) $P\left(\overline{A}\right) + P\left(\overline{B}\right)$

Q140. JEE Main 2018 (Online) 16th April Morning Slot

Two different families A and B are blessed with equal number of children. There are 3 tickets to be distributed amongst the children of these families so that no child gets more than one ticket. If the probability that all the tickets go to the children of the family B is $\frac{1}{12}$, then the number of children in each family is :

Open solution

(A) 3 (B) 4 (C) 5 (D) 6

Q141. JEE Main 2018 (Offline)

A bag contains 4 red and 6 black balls. A ball is drawn at random from the bag, its colour is observed and this ball along with two additional balls of the same colour are returned to the bag. If now a ball is drawn at random from the bag, then the probability that this drawn ball is red, is :

Open solution

(A) $\frac{3}{4}$ (B) $\frac{3}{10}$ (C) $\frac{2}{5}$ (D) $\frac{1}{5}$

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Q142. JEE Main 2018 (Online) 15th April Evening Slot

A player X has a biased coin whose probability of showing heads is p and a player Y has a fair coin. They start playing a game with their own coins and play alternately. The player who throws a head first is a winner. If X starts the game, and the probability of winning the game by both the players is equal, then the value of 'p' is :

Open solution

(A) $\frac{1}{5}$ (B) $\frac{1}{3}$ (C) $\frac{2}{5}$ (D) $\frac{1}{4}$

Q143. JEE Main 2018 (Online) 15th April Morning Slot

A box 'A' contains 2 white, 3 red and 2 black balls. Another box 'B' contains 4 white, 2 red and 3 black balls. If two balls are drawn at random, without replacement, from a randomly selected box and one ball turns out to be white while the other ball turns out to be red, then the probability that both balls are drawn from box 'B' is :

Open solution

(A) $\frac{9}{16}$ (B) $\frac{7}{16}$ (C) $\frac{9}{32}$ (D) $\frac{7}{8}$

Q144. JEE Main 2017 (Online) 9th April Morning Slot

From a group of 10 men and 5 women, four member committees are to be formed each of which must contain at least one woman. Then the probability for these committees to have more women than men, is :

Open solution

(A) $\frac{21}{220}$ (B) $\frac{3}{11}$ (C) $\frac{1}{11}$ (D) $\frac{2}{23}$

Q145. JEE Main 2017 (Online) 9th April Morning Slot

Let E and F be two independent events. The probability that both E and F happen is $\frac{1}{12}$ and the probability that neither E nor F happens is $\frac{1}{2}$, then a value of $\frac{P(E)}{P(F)}$ is :

Open solution

(A) $\frac{4}{3}$ (B) $\frac{3}{2}$ (C) $\frac{1}{3}$ (D) $\frac{5}{12}$

Q146. JEE Main 2017 (Online) 8th April Morning Slot

Three persons P, Q and R independently try to hit a target. I the probabilities of their hitting the target are $\frac{3}{4}$, $\frac{1}{2}$ and $\frac{5}{8}$ respectively, then the probability that the target is hit by P or Q but not by R is :

Open solution

(A) $\frac{21}{64}$ (B) $\frac{9}{64}$ (C) $\frac{15}{64}$ (D) $\frac{39}{64}$

Q147. JEE Main 2017 (Online) 8th April Morning Slot

An unbiased coin is tossed eight times. The probability of obtaining at least one head and at least one tail is :

Open solution

(A) $\frac{255}{256}$ (B) $\frac{127}{128}$ (C) $\frac{63}{64}$ (D) $\frac{1}{2}$

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Q148. JEE Main 2017 (Offline)

For three events A, B and C,

P(Exactly one of A or B occurs)
= P(Exactly one of B or C occurs)
= P (Exactly one of C or A occurs) = $\frac{1}{4}$
and P(All the three events occur simultaneously) = $\frac{1}{16}$.

Then the probability that at least one of the events occurs, is :

(A) $\frac{7}{16}$ (B) $\frac{7}{64}$ (C) $\frac{3}{16}$ (D) $\frac{7}{32}$

Open solution

Q149. JEE Main 2017 (Offline)

If two different numbers are taken from the set {0, 1, 2, 3, ..., 10}; then the probability that their sum as well as absolute difference are both multiple of 4, is :

(A) $\frac{12}{55}$ (B) $\frac{14}{45}$ (C) $\frac{7}{55}$ (D) $\frac{6}{55}$

Open solution

Q150. JEE Main 2016 (Online) 9th April Morning Slot

If A and B are any two events such that P(A) = $\frac{2}{5}$ and P (A ∩ B) = $\frac{3}{20}$, hen the conditional probability, P(A |(A' ∪ B')), where A' denotes the complement of A, is equal to :

(A) $\frac{1}{4}$ (B) $\frac{5}{17}$ (C) $\frac{8}{17}$ (D) $\frac{11}{20}$

Open solution

Q151. JEE Main 2016 (Offline)

Let two fair six-faced dice A and B be thrown simultaneously. If E_1 is the event that die A shows up four, E_2 is the event that die B shows up two and E_3 is the event that the sum of numbers on both dice is odd, then which of the following statements is **NOT** true?

(A) E_1 and E_2 are independent. (B) E_2 and E_3 are independent.
(C) E_1 and E_3 are independent. (D) E_1, E_2 and E_3 are independent.

Open solution

Q152. JEE Main 2015 (Online) 11th April Morning Slot

If the lengths of the sides of a triangle are decided by the three throws of a single fair die, then the probability that the triangle is of maximum area given that it is an isosceles triangle, is :

(A) $\frac{1}{26}$ (B) $\frac{1}{27}$ (C) $\frac{1}{21}$ (D) $\frac{1}{15}$

Open solution

Q153. JEE Main 2015 (Online) 10th April Morning Slot

Let X be a set containing 10 elements and P(X) be its power set. If A and B are picked up at random from P(X), with replacement, then the probability that A and B have equal number of elements, is :

(A) $\frac{^{20}C_{10}}{2^{10}}$ (B) $\frac{(2^{10}-1)}{2^{20}}$
(C) $\frac{(2^{10}-1)}{2^{10}}$ (D) $\frac{^{20}C_{10}}{2^{20}}$

Open solution

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Q154. JEE Main 2015 (Offline)

If 12 different balls are to be placed in 3 identical boxes, then the probability that one of the boxes contains exactly 3 balls is :

(A) $220\left(\frac{1}{3}\right)^{12}$ (B) $22\left(\frac{1}{3}\right)^{11}$

(C) $\frac{55}{3}\left(\frac{2}{3}\right)^{11}$ (D) $55\left(\frac{2}{3}\right)^{10}$

Open solution

Q155. JEE Main 2014 (Online) 19th April Morning Slot

Let A and E be any two events with positive probabilities :

Statement - 1 : P(E / A) $\geq$ P(A/E)P(E)

Statement - 2 : P(A/E) $\geq$ P(A$\cap$E).

(A) Both the statements are true

(B) Both the statements are false

(C) Statement - 1 is true, Statement - 2 is false

(D) Statement - 1 is false, statement - 2 is true

Open solution

Q156. JEE Main 2014 (Online) 12th April Morning Slot

A number x is chosen at random from the set {1, 2, 3, 4, ... , 100}. Define the event : A = the chosen number x satisfies $\frac{(x-10)(x-50)}{(x-30)} \geq 0$

Then P(A) is :

(A) 0.71 (B) 0.70 (C) 0.51 (D) 0.20

Open solution

Q157. JEE Main 2014 (Online) 11th April Morning Slot

A set S contains 7 elements. A non-empty subset A of S and an element x of S are chosen at random. Then the probability that x$\in$A is :

(A) $\frac{1}{2}$ (B) $\frac{64}{127}$ (C) $\frac{63}{128}$ (D) $\frac{31}{128}$

Open solution

Q158. JEE Main 2014 (Online) 9th April Morning Slot

If A and B are two events such that $P(A \cup B) = P(A \cap B)$, then the **incorrect** statement amongst the following statements is :

(A) A and B are equally likely (B) $P(A \cap B') = 0$

(C) $P(A' \cap B) = 0$ (D) $P(A) + P(B) = 1$

Open solution

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Q159. JEE Main 2014 (Offline)

Let A and B be two events such that $P\left(\overline{A \cup B}\right) = \frac{1}{6}$, $P(A \cap B) = \frac{1}{4}$ and $P\left(\overline{A}\right) = \frac{1}{4}$, where $\overline{A}$ stands for the complement of the event A. Then the events A and B are :

(A) independent but not equally likely.
(B) independent and equally likely.
(C) mutually exclusive and independent.
(D) equally likely but not independent.

Open solution

Q160. JEE Main 2013 (Online) 25th April Morning Slot

If the events A and B are mutually exclusive events such that P(A) = $\frac{3x+1}{3}$ and P(B) = $\frac{1-x}{4}$, then the set of possible values of x lies in the interval:

(A) [0, 1]
(B) $[\frac{1}{3}, \frac{2}{3}]$
(C) $[-\frac{1}{3}, \frac{5}{9}]$
(D) $[-\frac{7}{9}, \frac{4}{9}]$

Open solution

Q161. JEE Main 2013 (Online) 23th April Morning Slot

A, B, C try to hit a target simultaneously but independently. Their respective probabilities of hitting the targets are $\frac{3}{4}$, $\frac{1}{2}$, $\frac{5}{8}$. The probability that the target is hit by A or B but not by C is :

(A) 21/64
(B) 7/8
(C) 7/32
(D) 9/64

Open solution

Q162. JEE Main 2013 (Online) 22th April Morning Slot

Given two independent events, if the probability that exactly one of them occurs is $\frac{26}{49}$ and the probability that none of them occurs is $\frac{15}{49}$, then the probability of more probable of the two events is :

(A) 4/7
(B) 6/7
(C) 3/7
(D) 5/7

Open solution

Q163. AIEEE 2012 (Online) 26th May

There are two balls in an urn. Each ball can be either white or black. If a white ball is put into the urn and there after a ball is drawn at random from the urn, then the probability that it is white is

(A) $\frac{1}{4}$
(B) $\frac{2}{3}$
(C) $\frac{1}{5}$
(D) $\frac{1}{3}$

Open solution

Q164. AIEEE 2012 (Online) 19th May

If six students, including two particular students A and B, stand in a row, then the probability that A and B are separated with one student in between them is

(A) $\frac{8}{15}$
(B) $\frac{4}{15}$
(C) $\frac{2}{15}$
(D) $\frac{1}{15}$

Open solution

Q165. AIEEE 2012 (Online) 12th May

A number n is randomly selected from the set $\{1, 2, 3, \ldots, 1000\}$. The probability that $\frac{\sum_{i=1}^{n} i^2}{\sum_{i=1}^{n} i}$ is an integer is

(A) 0.331
(B) 0.333
(C) 0.334
(D) 0.332

Open solution

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Q166. AIEEE 2012 (Online) 7th May

Open solution

Let X and Y are two events such that $P(X \cup Y) = P(X \cap Y)$.
Statement 1: $P(X \cap Y') = P(X' \cap Y) = 0$
Statement 2: $P(X) + P(Y) = 2P(X \cap Y)$

(A) Statement 1 is false, Statement 2 is true.

(B) Statement 1 is true, Statement 2 is true, Statement 2 is not a correct explanation of Statement 1.

(C) Statement 1 is true, Statement 2 is false.

(D) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation of Statement 1.

Q167. AIEEE 2012

Open solution

Three numbers are chosen at random without replacement from $\{1, 2, 3, ..8\}$. The probability that their minimum is 3, given that their maximum is 6, is :

(A) $\frac{3}{8}$ (B) $\frac{1}{5}$ (C) $\frac{1}{4}$ (D) $\frac{2}{5}$

Q168. AIEEE 2011

Open solution

If C and D are two events such that $C \subset D$ and $P(D) \neq 0$, then the correct statement among the following is :

(A) $P\left(\frac{C}{D}\right) \geq P(C)$

(B) $P\left(\frac{C}{D}\right) < P(C)$

(C) $P\left(\frac{C}{D}\right) = \frac{P(D)}{P(C)}$

(D) $P\left(\frac{C}{D}\right) = P(C)$

Q169. AIEEE 2010

Open solution

Four numbers are chosen at random (without replacement) from the set $\{1, 2, 3, ...20\}$.

Statement - 1: The probability that the chosen numbers when arranged in some order will form an AP is $\frac{1}{85}$.
Statement - 2: If the four chosen numbers form an AP, then the set of all possible values of common difference is $(\pm 1, \pm 2, \pm 3, \pm 4, \pm 5)$.

(A) Statement - 1 is true, Statement - 2 is true; Statement - 2 is **not** a correct explanation for Statement - 1.

(B) Statement - 1 is true, Statement - 2 is false.

(C) Statement - 1 is false, Statement -2 is true.

(D) Statement - 1 is true, Statement - 2 is true; Statement - 2 is a correct explanation for Statement - 1.

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Q170. AIEEE 2010

An urn contains nine balls of which three are red, four are blue and two are green. Three balls are drawn at random without replacement from the urn. The probability that the three balls have different colours is :

(A) $\frac{2}{7}$ (B) $\frac{1}{21}$ (C) $\frac{1}{23}$ (D) $\frac{1}{3}$

Open solution

Q171. AIEEE 2009

One ticket is selected at random from 50 tickets numbered $00, 01, 02, \ldots, 49$. Then the probability that the sum of the digits on the selected ticket is 8, given that the product of these digits is zero, equals :

(A) $\frac{1}{7}$ (B) $\frac{5}{14}$ (C) $\frac{1}{50}$ (D) $\frac{1}{14}$

Open solution

Q172. AIEEE 2008

It is given that the events A and B are such that $P(A) = \frac{1}{4}, P(A|B) = \frac{1}{2}$ and $P(B|A) = \frac{2}{3}$. Then $P(B)$ is :

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{1}{2}$

Open solution

Q173. AIEEE 2008

A die is thrown. Let A be the event that the number obtained is greater than 3. Let B be the event that the number obtained is less than 5. Then $P(A \cup B)$ is :

(A) $\frac{3}{5}$ (B) 0 (C) 1 (D) $\frac{2}{5}$

Open solution

Q174. AIEEE 2007

Two aeroplanes I and II bomb a target in succession. The probabilities of I and II scoring a hit correctly are 0.3 and 0.2, respectively. The second plane will bomb only if the first misses the target. The probability that the target is hit by the second plane is :

(A) 0.2 (B) 0.7 (C) 0.06 (D) 0.32

Open solution

Q175. AIEEE 2006

At a telephone enquiry system the number of phone calls regarding relevant enquiry follow Poisson distribution with an average of 5 phone calls during 10 minute time intervals. The probability that there is at the most one phone call during a 10-minute time period is :

(A) $\frac{6}{5^e}$ (B) $\frac{5}{6}$ (C) $\frac{6}{55}$ (D) $\frac{6}{e^5}$

Open solution

Q176. AIEEE 2005

Three houses are available in a locality. Three persons apply for the houses. Each applies for one house without consulting others. The probability that all the three apply for the same house is :

(A) $\frac{2}{9}$ (B) $\frac{1}{9}$ (C) $\frac{8}{9}$ (D) $\frac{7}{9}$

Open solution

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Q177. AIEEE 2005

A random variable X has Poisson distribution with mean 2.
Then $P(X > 1.5)$ equals :

(A) $\frac{2}{e^2}$ (B) 0 (C) $1 - \frac{3}{e^2}$ (D) $\frac{3}{e^2}$

Open solution

Q178. AIEEE 2005

Let A and B two events such that $P\left(\overline{A \cup B}\right) = \frac{1}{6}$, $P(A \cap B) = \frac{1}{4}$ and $P\left(\overline{A}\right) = \frac{1}{4}$, where $\overline{A}$ stands for complement of event A. Then events A and B are :

(A) equally likely and mutually exclusive
(B) equally likely but not independent
(C) independent but not equally likely
(D) mutually exclusive and independent

Open solution

Q179. AIEEE 2004

The probability that A speaks truth is $\frac{4}{5}$, while the probability for B is $\frac{3}{4}$. The probability that they contradict each other when asked to speak on a fact is :

(A) $\frac{4}{5}$ (B) $\frac{1}{5}$ (C) $\frac{7}{20}$ (D) $\frac{3}{20}$

Open solution

Q180. AIEEE 2003

Events A, B, C are mutually exclusive events such that $P(A) = \frac{3x+1}{3}$, $P(B) = \frac{1-x}{4}$ and $P(C) = \frac{1-2x}{2}$. The set of possible values of x are in the interval.

(A) $[0, 1]$
(B) $\left[\frac{1}{3}, \frac{1}{2}\right]$
(C) $\left[\frac{1}{3}, \frac{2}{3}\right]$
(D) $\left[\frac{1}{3}, \frac{13}{3}\right]$

Open solution

Q181. AIEEE 2003

Five horses are in a race. Mr. A selects two of the horses at random and bets on them. The probability that Mr. A selected the winning horse is :

(A) $\frac{2}{5}$ (B) $\frac{4}{5}$ (C) $\frac{3}{5}$ (D) $\frac{1}{5}$

Open solution

Q182. AIEEE 2002

A problem in mathematics is given to three students A, B, C and their respective probability of solving the problem is $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$. Probability that the problem is solved is :

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{1}{3}$

Open solution

Q183. AIEEE 2002

A and B are events such that $P(A \cup B) = 3/4$, $P(A \cap B) = 1/4$, $P\left(\overline{A}\right) = 2/3$ then $P\left(\overline{A} \cap B\right)$ is :

(A) $5/12$ (B) $3/8$ (C) $5/8$ (D) $1/4$

Open solution

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Numerical 33 questions

Q184. JEE Main 2026 (Online) 4th April Evening Shift

From a month of 31 days, 3 different dates are selected at random. If the probability that these dates are in an increasing A.P. is equal to $\frac{a}{b}$, where $a, b \in$ N and $\gcd(a, b) = 1$, then $a + b$ is equal to ____

Open solution

Q185. JEE Main 2026 (Online) 4th April Morning Shift

A coin is tossed 8 times. If the probability that exactly 4 heads appear in the first six tosses and exactly 3 heads appear in the last five tosses is p, then $96p$ is equal to ____ .

Open solution

Q186. JEE Main 2026 (Online) 2nd April Morning Shift

Let **a, b, c** ∈ {1, 2, 3, 4}. If the probability, that $ax^2 + 2\sqrt{2}bx + c > 0$ for all $x \in \mathbb{R}$, is $\frac{m}{n}$, $\gcd(m, n) = 1$, then $m + n$ is equal to _____.

Open solution

Q187. JEE Main 2026 (Online) 24th January Evening Shift

Let S be a set of 5 elements and $\mathrm{P}(\mathrm{S})$ denote the power set of S . Let E be an event of choosing an ordered pair (A, B) from the set $\mathrm{P}(\mathrm{S}) \times \mathrm{P}(\mathrm{S})$ such that $\mathrm{A} \cap \mathrm{B} = \emptyset$. If the probability of the event E is $\frac{3^p}{2^q}$, where $p, q \in N$, then $p + q$ is equal to

Open solution

Q188. JEE Main 2026 (Online) 23rd January Morning Shift

From the first 100 natural numbers, two numbers first a and then b are selected randomly without replacement. If the probability that $\mathrm{a} - \mathrm{b} \geqslant 10$ is $\frac{\mathrm{m}}{\mathrm{n}}$, $\gcd(\mathrm{m}, \mathrm{n}) = 1$, then $\mathrm{m} + \mathrm{n}$ is equal to ____ .

Open solution

Q189. JEE Main 2025 (Online) 4th April Evening Shift

A card from a pack of 52 cards is lost. From the remaining 51 cards, n cards are drawn and are found to be spades. If the probability of the lost card to be a spade is $\frac{11}{50}$, then n is equal to _____ .

Open solution

Q190. JEE Main 2025 (Online) 2nd April Morning Shift

Three distinct numbers are selected randomly from the set $\{1, 2, 3, \ldots, 40\}$. If the probability, that the selected numbers are in an increasing G.P., is $\frac{m}{n}$, $\gcd(m, n) = 1$, then $m + n$ is equal to _____ .

Open solution

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Q191. JEE Main 2024 (Online) 9th April Morning Shift

Let a, b and c denote the outcome of three independent rolls of a fair tetrahedral die, whose four faces are marked $1, 2, 3, 4$. If the probability that $ax^2 + bx + c = 0$ has all real roots is $\frac{m}{n}, \gcd(\mathrm{m}, \mathrm{n}) = 1$, then $\mathrm{m} + \mathrm{n}$ is equal to _____.

Open solution

Q192. JEE Main 2024 (Online) 8th April Morning Shift

Three balls are drawn at random from a bag containing 5 blue and 4 yellow balls. Let the random variables X and Y respectively denote the number of blue and yellow balls. If $\bar{X}$ and $\bar{Y}$ are the means of X and Y respectively, then $7\bar{X} + 4\bar{Y}$ is equal to _____.

Open solution

Q193. JEE Main 2024 (Online) 6th April Evening Shift

From a lot of 12 items containing 3 defectives, a sample of 5 items is drawn at random. Let the random variable X denote the number of defective items in the sample. Let items in the sample be drawn one by one without replacement. If variance of X is $\frac{m}{n}$, where $\gcd(m, n) = 1$, then $n - m$ is equal to _____.

Open solution

Q194. JEE Main 2024 (Online) 5th April Morning Shift

From a lot of 10 items, which include 3 defective items, a sample of 5 items is drawn at random. Let the random variable X denote the number of defective items in the sample. If the variance of X is σ^2, then $96\sigma^2$ is equal to _____.

Open solution

Q195. JEE Main 2024 (Online) 4th April Evening Shift

In a tournament, a team plays 10 matches with probabilities of winning and losing each match as $\frac{1}{3}$ and $\frac{2}{3}$ respectively. Let x be the number of matches that the team wins, and y be the number of matches that team loses. If the probability $\mathrm{P}(|x - y| \leq 2)$ is p, then $3^9 p$ equals _____.

Open solution

Q196. JEE Main 2024 (Online) 30th January Morning Shift

A group of 40 students appeared in an examination of 3 subjects - Mathematics, Physics and Chemistry. It was found that all students passed in at least one of the subjects, 20 students passed in Mathematics, 25 students passed in Physics, 16 students passed in Chemistry, at most 11 students passed in both Mathematics and Physics, at most 15 students passed in both Physics and Chemistry, at most 15 students passed in both Mathematics and Chemistry. The maximum number of students passed in all the three subjects is _____.

Open solution

Q197. JEE Main 2024 (Online) 27th January Morning Shift

A fair die is tossed repeatedly until a six is obtained. Let X denote the number of tosses required and let

$a = P(X = 3), b = P(X \geqslant 3)$ and $c = P(X \geqslant 6 \mid X > 3)$. Then $\frac{b+c}{a}$ is equal to _____.

Open solution

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Q198. JEE Main 2023 (Online) 12th April Morning Shift

A fair $n(n > 1)$ faces die is rolled repeatedly until a number less than n appears. If the mean of the number of tosses required is $\frac{n}{9}$, then n is equal to _____.

Open solution

Q199. JEE Main 2023 (Online) 11th April Evening Shift

Let the probability of getting head for a biased coin be $\frac{1}{4}$. It is tossed repeatedly until a head appears. Let N be the number of tosses required. If the probability that the equation $64\mathrm{x}^2 + 5\mathrm{Nx} + 1 = 0$ has no real root is $\frac{\mathrm{p}}{\mathrm{q}}$, where p and q are coprime, then $q - p$ is equal to _____.

Open solution

Q200. JEE Main 2023 (Online) 31st January Evening Shift

Let A be the event that the absolute difference between two randomly chosen real numbers in the sample space $[0, 60]$ is less than or equal to a . If $\mathrm{P(A)} = \frac{11}{36}$, then a is equal to _____.

Open solution

Q201. JEE Main 2023 (Online) 30th January Evening Shift

A bag contains six balls of different colours. Two balls are drawn in succession with replacement. The probability that both the balls are of the same colour is p. Next four balls are drawn in succession with replacement and the probability that exactly three balls are of the same colour is q. If $p : q = m : n$, where m and n are coprime, then $m + n$ is equal to :

Open solution

Q202. JEE Main 2023 (Online) 25th January Evening Shift

25% of the population are smokers. A smoker has 27 times more chances to develop lung cancer than a non smoker. A person is diagnosed with lung cancer and the probability that this person is a smoker is $\frac{k}{10}$. Then the value of k is _____.

Open solution

Q203. JEE Main 2023 (Online) 24th January Evening Shift

Three urns A, B and C contain 4 red, 6 black; 5 red, 5 black; and λ red, 4 black balls respectively. One of the urns is selected at random and a ball is drawn. If the ball drawn is red and the probability that it is drawn from urn C is 0.4 then the square of the length of the side of the largest equilateral triangle, inscribed in the parabola $y^2 = \lambda x$ with one vertex at the vertex of the parabola, is :

Open solution

Q204. JEE Main 2022 (Online) 30th July Evening Shift

The probability distribution of a random variable X is $P(X = i) = \frac{1}{2^i}$, $i = 1, 2, 3, \ldots$. The variance of X is equal to _____.

Open solution

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Q205. JEE Main 2022 (Online) 28th July Evening Shift

A bag contains 4 white and 6 black balls. Three balls are drawn at random from the bag. Let X be the number of white balls, among the drawn balls. If σ^2 is the variance of X, then $100\sigma^2$ is equal to _____.

Open solution

Q206. JEE Main 2022 (Online) 30th June Morning Shift

The probability distribution of X is :

X	0	1	2	3
P(X)	$\frac{1-d}{4}$	$\frac{1+2d}{4}$	$\frac{1-4d}{4}$	$\frac{1+3d}{4}$

For the minimum possible value of d, sixty times the mean of X is equal to _____.

Open solution

Q207. JEE Main 2022 (Online) 27th June Evening Shift

Let S = {E_1, E_2, ..., E_8} be a sample space of a random experiment such that $P(E_n) = \frac{n}{36}$ for every n = 1, 2, ..., 8. Then the number of elements in the set $\left\{A \subseteq S : P(A) \geq \frac{4}{5}\right\}$ is _____.

Open solution

Q208. JEE Main 2022 (Online) 26th June Evening Shift

If the probability that a randomly chosen 6-digit number formed by using digits 1 and 8 only is a multiple of 21 is p, then 96 p is equal to _____.

Open solution

Q209. JEE Main 2021 (Online) 1st September Evening Shift

Let X be a random variable with distribution.

x	-2	-1	3	4	6
P(X = x)	$\frac{1}{5}$	a	$\frac{1}{3}$	$\frac{1}{5}$	b

If the mean of X is 2.3 and variance of X is σ^2, then 100 σ^2 is equal to :

Open solution

Q210. JEE Main 2021 (Online) 31st August Morning Shift

An electric instrument consists of two units. Each unit must function independently for the instrument to operate. The probability that the first unit functions is 0.9 and that of the second unit is 0.8. The instrument is switched on and it fails to operate. If the probability that only the first unit failed and second unit is functioning is p, then 98 p is equal to _____.

Open solution

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Q211. JEE Main 2021 (Online) 27th August Evening Shift

The probability distribution of random variable X is given by :

X	1	2	3	4	5
P(X)	K	2K	2K	3K	K

Let p = P(1 < X < 4 | X < 3). If 5p = λK, then λ equal to _____.

Open solution

Q212. JEE Main 2021 (Online) 25th July Evening Shift

A fair coin is tossed n-times such that the probability of getting at least one head is at least 0.9. Then the minimum value of n is _____.

Open solution

Q213. JEE Main 2021 (Online) 17th March Morning Shift

Let there be three independent events E_1, E_2 and E_3. The probability that only E_1 occurs is α, only E_2 occurs is β and only E_3 occurs is γ. Let 'p' denote the probability of none of events occurs that satisfies the equations
$(\alpha - 2\beta)p = \alpha\beta$ and $(\beta - 3\gamma)p = 2\beta\gamma$. All the given probabilities are assumed to lie in the interval (0, 1).

Then, $\frac{Probability\ of\ occurrence\ of\ E_1}{Probability\ of\ occurrence\ of\ E_3}$ is equal to _____.

Open solution

Q214. JEE Main 2021 (Online) 24th February Morning Shift

Let B_i (i = 1, 2, 3) be three independent events in a sample space. The probability that only B_1 occur is α, only B_2 occurs is β and only B_3 occurs is γ. Let p be the probability that none of the events B_i occurs and these 4 probabilities satisfy the equations $(\alpha - 2\beta)p = \alpha\beta$ and $(\beta - 3\gamma)p = 2\beta\gamma$ (All the probabilities are assumed to lie in the interval (0, 1)).
Then $\frac{P(B_1)}{P(B_3)}$ is equal to _____.

Open solution

Q215. JEE Main 2020 (Online) 5th September Morning Slot

Four fair dice are thrown independently 27 times. Then the expected number of times, at least two dice show up a three or a five, is _____.

Open solution

Q216. JEE Main 2020 (Online) 4th September Morning Slot

The probability of a man hitting a target is $\frac{1}{10}$. The least number of shots required, so that the probability of his hitting the target at least once is greater than $\frac{1}{4}$, is _____.

Open solution

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Out of syllabus 38 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q217. JEE Main 2023 (Online) 13th April Evening Shift · MCQ (Single Correct Answer)

The random variable X follows binomial distribution $\mathrm{B(n, p)}$, for which the difference of the mean and the variance is 1 . If $2\mathrm{P(X=2)} = 3\mathrm{P(X=1)}$, then $n^2\mathrm{P(X>1)}$ is equal to :

(A) 15 (B) 12 (C) 11 (D) 16

Open solution

Q218. JEE Main 2023 (Online) 10th April Evening Shift · MCQ (Single Correct Answer)

Let a die be rolled n times. Let the probability of getting odd numbers seven times be equal to the probability of getting odd numbers nine times. If the probability of getting even numbers twice is $\frac{k}{2^{15}}$, then k is equal to :

(A) 15 (B) 60 (C) 30 (D) 90

Open solution

Q219. JEE Main 2023 (Online) 6th April Morning Shift · MCQ (Single Correct Answer)

A pair of dice is thrown 5 times. For each throw, a total of 5 is considered a success. If the probability of at least 4 successes is $\frac{k}{3^{11}}$, then k is equal to :

(A) 82 (B) 164 (C) 123 (D) 75

Open solution

Q220. JEE Main 2023 (Online) 1st February Morning Shift · MCQ (Single Correct Answer)

In a binomial distribution $B(n, p)$, the sum and the product of the mean and the variance are 5 and 6 respectively, then $6(n + p - q)$ is equal to :

(A) 52 (B) 50 (C) 51 (D) 53

Open solution

Q221. JEE Main 2022 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

Let X have a binomial distribution B(n, p) such that the sum and the product of the mean and variance of X are 24 and 128 respectively. If $P(X > n - 3) = \frac{k}{2^n}$, then k is equal to :

(A) 528 (B) 529 (C) 629 (D) 630

Open solution

Q222. JEE Main 2022 (Online) 26th July Evening Shift · MCQ (Single Correct Answer)

Let X be a binomially distributed random variable with mean 4 and variance $\frac{4}{3}$. Then, $54\,P(X \leq 2)$ is equal to :

(A) $\frac{73}{27}$ (B) $\frac{146}{27}$ (C) $\frac{146}{81}$ (D) $\frac{126}{81}$

Open solution

Q223. JEE Main 2022 (Online) 26th July Morning Shift · MCQ (Single Correct Answer)

The mean and variance of a binomial distribution are α and $\frac{\alpha}{3}$ respectively. If $\mathrm{P}(X = 1) = \frac{4}{243}$, then $\mathrm{P}(X = 4$ or $5)$ is equal to :

(A) $\frac{5}{9}$ (B) $\frac{64}{81}$ (C) $\frac{16}{27}$ (D) $\frac{145}{243}$

Open solution

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Q224. JEE Main 2022 (Online) 25th July Morning Shift · MCQ (Single Correct Answer)

If the sum and the product of mean and variance of a binomial distribution are 24 and 128 respectively, then the probability of one or two successes is :

(A) $\frac{33}{2^{32}}$ (B) $\frac{33}{2^{29}}$ (C) $\frac{33}{2^{28}}$ (D) $\frac{33}{2^{27}}$

Open solution

Q225. JEE Main 2022 (Online) 30th June Morning Shift · MCQ (Single Correct Answer)

If a random variable X follows the Binomial distribution B(5, p) such that P(X = 0) = P(X = 1), then $\frac{P(X=2)}{P(X=3)}$ is equal to :

(A) 1 (B) 10 (C) 25 (D) 5

Open solution

Q226. JEE Main 2022 (Online) 27th June Morning Shift · MCQ (Single Correct Answer)

Let X be a random variable having binomial distribution B(7, p). If P(X = 3) = 5P(X = 4), then the sum of the mean and the variance of X is :

(A) $\frac{105}{16}$ (B) $\frac{7}{16}$ (C) $\frac{77}{36}$ (D) $\frac{49}{16}$

Open solution

Q227. JEE Main 2022 (Online) 26th June Morning Shift · MCQ (Single Correct Answer)

Let a biased coin be tossed 5 times. If the probability of getting 4 heads is equal to the probability of getting 5 heads, then the probability of getting atmost two heads is :

(A) $\frac{275}{6^5}$ (B) $\frac{36}{5^4}$ (C) $\frac{181}{5^5}$ (D) $\frac{46}{6^4}$

Open solution

Q228. JEE Main 2022 (Online) 24th June Morning Shift · MCQ (Single Correct Answer)

If a random variable X follows the Binomial distribution B(33, p) such that

$3P(X = 0) = P(X = 1)$, then the value of $\frac{P(X=15)}{P(X=18)} - \frac{P(X=16)}{P(X=17)}$ is equal to :

(A) 1320 (B) 1088 (C) $\frac{120}{1331}$ (D) $\frac{1088}{1089}$

Open solution

Q229. JEE Main 2021 (Online) 27th August Evening Shift · MCQ (Single Correct Answer)

Each of the persons A and B independently tosses three fair coins. The probability that both of them get the same number of heads is :

(A) $\frac{1}{8}$ (B) $\frac{5}{8}$ (C) $\frac{5}{16}$ (D) 1

Open solution

Q230. JEE Main 2021 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

A student appeared in an examination consisting of 8 true-false type questions. The student guesses the answers with equal probability. the smallest value of n, so that the probability of guessing at least 'n' correct answers is less than $\frac{1}{2}$, is :

(A) 5 (B) 6 (C) 3 (D) 4

Open solution

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Q231. JEE Main 2021 (Online) 18th March Evening Shift · MCQ (Single Correct Answer)

Let in a Binomial distribution, consisting of 5 independent trials, probabilities of exactly 1 and 2 successes be 0.4096 and 0.2048 respectively. Then the probability of getting exactly 3 successes is equal to :

(A) $\frac{40}{243}$ (B) $\frac{128}{625}$ (C) $\frac{80}{243}$ (D) $\frac{32}{625}$

Open solution

Q232. JEE Main 2021 (Online) 26th February Morning Shift · MCQ (Single Correct Answer)

A fair coin is tossed a fixed number of times. If the probability of getting 7 heads is equal to probability of getting 9 heads, then the probability of getting 2 heads is :

(A) $\frac{15}{2^8}$ (B) $\frac{15}{2^{12}}$ (C) $\frac{15}{2^{13}}$ (D) $\frac{15}{2^{14}}$

Open solution

Q233. JEE Main 2021 (Online) 24th February Morning Shift · MCQ (Single Correct Answer)

An ordinary dice is rolled for a certain number of times. If the probability of getting an odd number 2 times is equal to the probability of getting an even number 3 times, then the probability of getting an odd number for odd number of times is :

(A) $\frac{5}{36}$ (B) $\frac{3}{16}$ (C) $\frac{1}{2}$ (D) $\frac{1}{32}$

Open solution

Q234. JEE Main 2020 (Online) 7th January Evening Slot · MCQ (Single Correct Answer)

In a workshop, there are five machines and the probability of any one of them to be out of service on a day is $\frac{1}{4}$. If the probability that at most two machines will be out of service on the same day is $\left(\frac{3}{4}\right)^3 k$, then k is equal to :

(A) $\frac{17}{4}$ (B) $\frac{17}{2}$ (C) $\frac{17}{8}$ (D) 4

Open solution

Q235. JEE Main 2019 (Online) 12th April Evening Slot · MCQ (Single Correct Answer)

For an initial screening of an admission test, a candidate is given fifty problems to solve. If the probability that the candidate solve any problem is $\frac{4}{5}$, then the probability that he is unable to solve less than two problems is :

(A) $\frac{164}{25}\left(\frac{1}{5}\right)^{48}$ (B) $\frac{316}{25}\left(\frac{4}{5}\right)^{48}$

(C) $\frac{201}{5}\left(\frac{1}{5}\right)^{49}$ (D) $\frac{54}{5}\left(\frac{4}{5}\right)^{49}$

Open solution

Q236. JEE Main 2019 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

Let a random variable X have a binomial distribution with mean 8 and variance 4. If $P(X \leq 2) = \frac{k}{2^{16}}$, then k is equal to :

(A) 17 (B) 1 (C) 137 (D) 121

Open solution

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Q237. JEE Main 2019 (Online) 11th January Evening Slot · MCQ (Single Correct Answer)

A bag contains 30 white balls and 10 red balls. 16 balls are drawn one by one randomly from the bag with replacement. If X be the number of white balls drawn, then $\left(\frac{mean\ of\ X}{standard\ deviation\ of\ X}\right)$ is equal to :

Open solution

(A) 4 (B) $3\sqrt{2}$ (C) $\frac{4\sqrt{3}}{3}$ (D) $4\sqrt{3}$

Q238. JEE Main 2019 (Online) 10th January Evening Slot · MCQ (Single Correct Answer)

If the probability of hitting a target by a shooter, in any shot, is $\frac{1}{3}$, then the minimum number of independent shots at the target required by him so that the probability of hitting the target atleast once is greater than $\frac{5}{6}$ is :

Open solution

(A) 4 (B) 6 (C) 5 (D) 3

Q239. JEE Main 2019 (Online) 9th January Morning Slot · MCQ (Single Correct Answer)

Two cards are drawn successively with replacement from a well-shuffled deck of 52 cards. Let X denote the random variable of number of aces obtained in the two drawn cards. Then P(X = 1) + P (X = 2) equals :

Open solution

(A) $\frac{25}{169}$ (B) $\frac{49}{169}$ (C) $\frac{24}{169}$ (D) $\frac{52}{169}$

Q240. JEE Main 2017 (Offline) · MCQ (Single Correct Answer)

A box contains 15 green and 10 yellow balls. If 10 balls are randomly drawn, one-by-one, with replacement, then the variance of the number of green balls drawn is :

Open solution

(A) 6 (B) 4 (C) $\frac{6}{25}$ (D) $\frac{12}{5}$

Q241. JEE Main 2016 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

An experiment succeeds twice as often as it fails. The probability of at least 5 successes in the six trials of this experiment is :

Open solution

(A) $\frac{240}{729}$ (B) $\frac{192}{729}$ (C) $\frac{256}{729}$ (D) $\frac{496}{729}$

Q242. JEE Main 2015 (Online) 11th April Morning Slot · MCQ (Single Correct Answer)

If the mean and the variance of a binomial variate X are 2 and 1 respectively, then the probability that X takes a value greater than or equal to one is :

Open solution

(A) $\frac{1}{16}$ (B) $\frac{9}{16}$ (C) $\frac{3}{4}$ (D) $\frac{15}{16}$

Q243. JEE Main 2014 (Online) 11th April Morning Slot · MCQ (Single Correct Answer)

If X has a binomial distribution, B(n, p) with parameters n and p such that P(X = 2) = P(X = 3), then E(X), the mean of variable X, is :

Open solution

(A) 2 − p (B) 3 − p (C) $\frac{p}{2}$ (D) $\frac{p}{3}$

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Q244. JEE Main 2013 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

The probability of a man hitting a target is $\frac{2}{5}$. He fires at the target k times (k, a given number). Then the minimum k, so that the probability of hitting the target at least once is more than $\frac{7}{10}$, is :

(A) 3 (B) 5 (C) 2 (D) 4

Open solution

Q245. JEE Main 2013 (Offline) · MCQ (Single Correct Answer)

A multiple choice examination has 5 questions. Each question has three alternative answers of which exactly one is correct. The probability that a student will get 4 or more correct answers just by guessing is :

(A) $\frac{17}{3^5}$ (B) $\frac{13}{3^5}$ (C) $\frac{11}{3^5}$ (D) $\frac{10}{3^5}$

Open solution

Q246. AIEEE 2011 · MCQ (Single Correct Answer)

Consider 5 independent Bernoulli's trials each with probability of success p. If the probability of at least one failure is greater than or equal to $\frac{31}{32}$, then p lies in the interval :

(A) $\left(\frac{3}{4}, \frac{11}{12}\right]$ (B) $\left[0, \frac{1}{2}\right]$
(C) $\left(\frac{11}{12}, 1\right]$ (D) $\left(\frac{1}{2}, \frac{3}{4}\right]$

Open solution

Q247. AIEEE 2009 · MCQ (Single Correct Answer)

In a binomial distribution $B\left(n, p = \frac{1}{4}\right)$, if the probability of at least one success is greater than or equal to $\frac{9}{10}$, then n is greater than :

(A) $\frac{1}{\log_{10}^4 + \log_{10}^3}$ (B) $\frac{9}{\log_{10}^4 - \log_{10}^3}$
(C) $\frac{4}{\log_{10}^4 - \log_{10}^3}$ (D) $\frac{1}{\log_{10}^4 - \log_{10}^3}$

Open solution

Q248. AIEEE 2007 · MCQ (Single Correct Answer)

A pair of fair dice is thrown independently three times. The probability of getting a score of exactly 9 twice is :

(A) $8/729$ (B) $8/243$ (C) $1/729$ (D) $8/9$.

Open solution

Q249. AIEEE 2004 · MCQ (Single Correct Answer)

The mean and the variance of a binomial distribution are 4 and 2 respectively. Then the probability of 2 successes is :

(A) $\frac{28}{256}$ (B) $\frac{219}{256}$ (C) $\frac{128}{256}$ (D) $\frac{37}{256}$

Open solution

Q250. AIEEE 2003 · MCQ (Single Correct Answer)

The mean and variance of a random variable X having binomial distribution are 4 and 2 respectively, then $P(X = 1)$ is :

(A) $\frac{1}{4}$ (B) $\frac{1}{32}$ (C) $\frac{1}{16}$ (D) $\frac{1}{8}$

Open solution

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Q251. AIEEE 2002 · MCQ (Single Correct Answer)

A dice is tossed 5 times. Getting an odd number is considered a success. Then the variance of distribution of success is :

(A) $8/3$ (B) $3/8$ (C) $4/5$ (D) $5/4$

Open solution

Q252. JEE Main 2022 (Online) 29th July Evening Shift · Numerical

The sum and product of the mean and variance of a binomial distribution are 82.5 and 1350 respectively. Then the number of trials in the binomial distribution is _____.

Open solution

Q253. JEE Main 2022 (Online) 24th June Evening Shift · Numerical

In an examination, there are 10 true-false type questions. Out of 10, a student can guess the answer of 4 questions correctly with probability $\frac{3}{4}$ and the remaining 6 questions correctly with probability $\frac{1}{4}$. If the probability that the student guesses the answers of exactly 8 questions correctly out of 10 is $\frac{27k}{4^{10}}$, then k is equal to _____.

Open solution

Q254. JEE Main 2020 (Online) 5th September Evening Slot · Numerical

In a bombing attack, there is 50% chance that a bomb will hit the target. At least two independent hits are required to destroy the target completely. Then the minimum number of bombs, that must be dropped to ensure that there is at least 99% chance of completely destroying the target, is _____.

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **C**	3. **B**	4. **B**	5. **B**
6. **C**	7. **D**	8. **D**	9. **A**	10. **C**
11. **C**	12. **D**	13. **B**	14. **A**	15. **C**
16. **B**	17. **A**	18. **A**	19. **D**	20. **A**
21. **C**	22. **B**	23. **B**	24. **B**	25. **A**
26. **D**	27. **A**	28. **B**	29. **B**	30. **D**
31. **C**	32. **D**	33. **B**	34. **A**	35. **B**
36. **A**	37. **B**	38. **A**	39. **A**	40. **B**
41. **B**	42. **C**	43. **D**	44. **B**	45. **B**
46. **B**	47. **D**	48. **B**	49. **D**	50. **D**
51. **A**	52. **C**	53. **C**	54. **C**	55. **A**
56. **C**	57. **C**	58. **C**	59. **C**	60. **B**
61. **D**	62. **C**	63. **B**	64. **D**	65. **A**
66. **D**	67. **D**	68. **A**	69. **C**	70. **B**
71. **B**	72. **D**	73. **A**	74. **A**	75. **D**
76. **C**	77. **B**	78. **B**	79. **A**	80. **C**
81. **D**	82. **A**	83. **B**	84. **C**	85. **D**
86. **C**	87. **A**	88. **C**	89. **B**	90. **A**
91. **B**	92. **D**	93. **B**	94. **D**	95. **C**
96. **B**	97. **A**	98. **D**	99. **B**	100. **B**
101. **B**	102. **C**	103. **D**	104. **A**	105. **A**
106. **D**	107. **D**	108. **D**	109. **D**	110. **B**
111. **A**	112. **C**	113. **B**	114. **D**	115. **D**
116. **B**	117. **A**	118. **A**	119. **D**	120. **D**
121. **B**	122. **D**	123. **D**	124. **C**	125. **C**
126. **A**	127. **D**	128. **C**	129. **A**	130. **C**
131. **D**	132. **B**	133. **B**	134. **D**	135. **A**
136. **A**	137. **D**	138. **D**	139. **A**	140. **C**
141. **C**	142. **B**	143. **B**	144. **C**	145. **A**
146. **A**	147. **B**	148. **A**	149. **D**	150. **B**

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151. **D**	152. **B**	153. **D**	154. **C**	155. **A**
156. **A**	157. **B**	158. **D**	159. **A**	160. **C**
161. **A**	162. **A**	163. **B**	164. **B**	165. **C**
166. **B**	167. **B**	168. **A**	169. **B**	170. **A**
171. **D**	172. **B**	173. **C**	174. **D**	175. **D**
176. **B**	177. **C**	178. **C**	179. **C**	180. **B**
181. **A**	182. **A**	183. **A**		

Numerical

184. **944**	185. **9**	186. **81**	187. **15**	188. **311**
189. **2**	190. **2477**	191. **19**	192. **17**	193. **71**
194. **56**	195. **8288**	196. **10**	197. **12**	198. **10**
199. **27**	200. **10**	201. **14**	202. **9**	203. **432**
204. **2**	205. **57**	206. **75**	207. **19**	208. **33**
209. **781**	210. **28**	211. **30**	212. **4**	213. **6**
214. **6**	215. **11**	216. **3**		

Out of syllabus

217. **C**	218. **B**	219. **C**	220. **A**	221. **B**
222. **B**	223. **C**	224. **C**	225. **D**	226. **C**
227. **D**	228. **A**	229. **C**	230. **A**	231. **D**
232. **C**	233. **C**	234. **C**	235. **D**	236. **C**
237. **D**	238. **C**	239. **A**	240. **D**	241. **C**
242. **D**	243. **B**	244. **A**	245. **C**	246. **B**
247. **D**	248. **B**	249. **A**	250. **B**	251. **D**
252. **96**	253. **479**	254. **11**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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